

AERO

Tims gouge- if you can't pass DOR from life right now

- 1) Thrust available varies with RPM, ALT, and Temp
- 2) T-45 –tracks away from the crosswind
- 3) Max G below 5,000- +6.5 – (-3.0)
- 4) SBI causes a nose up pitch across all A/S- False
- 5) Rudder shock stabilization strips- reduce rudder buzz
- 6) Artificial stall warning ...- occurs at 21.5 AOA
- 7) T-45 does not incorporate- Area Rule
- 8) Spin recovery upright and inverted differ with – aileron position
- 9) Landing at a high elevation field- Same IAS but longer landing roll due to inc. TAS
- 10) -60 AOA A/S- (50-120 kts)
- 11) $\frac{3}{4}$ lateral stick- .92 - .95 Mach
- 12) Rudder full opposite- recovery both erect and inverted spins
- 13) Compared to normal approach, the T-45 Emergency Flap configuration has same approach A/S and faster stall A/S...
- 14) Approach Configuration stall buffet- 1 kt above stall
- 15) T/O distance (PA- 2,000 ft; Temp- 90 degrees F; 13,000 lbs weight; wind 10 kt headwind; half flap) – 2700 ft (don't need 50 ft obstacle clearance)
- 16) Stall AOA clean configuration- 26 AOA
- 17) 45 degree AOA ROD spin- 15,000 fpm
- 18) First shock wave .85- is the Force divergent Mach number
- 19) (1100 pds of fuel; 150 nm; 10,500 pds weight; calm wind) – (300, 220, 80, 500)
- 20) Rudder shakers- shake both pedals in both cockpits
- 21) Dutch roll- A/S below 220 KIAS (Tims considers that slow flight)
- 22) Transonic Flight (.8)- Mach Critical
- 23) Swept wing- stall wing tip first and have poor lateral stability
- 24) Approach configuration stall- 29-30 AOA
- 25) Tuck under- caused by aft movement of AC
- 26) T-45 high slip will cause rudder blowout and 250 lbs force...
- 27) External stores- decrease stall I AOA lower in cruise and approach configurations

- 28) T-45 upright spin NOT characterized by – recovery AOA increasing past 5 units
- 29) High speeds and low speeds characteristics improved by- vortex generators
- 30) Spiral divergence in clean configuration is possible- (False, need asymmetric load)
- 31) Flap extension plane will- balloon, settle, stability reduction
- 32) Pulling out of a dive decelerating .95 -.85 T-45 will pitch up- (True)
- 33) Max abort speed- commence stop on remaining runway without using over run or arresting gear.
- 34) Mach number- ratio of TAS/SOS
- 35) Two pilots shift CG forward and more stable
- 36) Heavy wing- $\frac{3}{4}$ lateral stick
- 37) Optimum AOA- 17 units
- 38) CG location and stability effected by – Consumption of fuel/ raising lowering landing gear
- 39) Disadvantage swept wing- stalls high AOA with Coefficient of lift reduced
- 40) Inverted spin- 0 AOA, neg G., turn needle in direction of spin

Engineering Timms Gouge

- Know limits chart and what would be an overtemp, max egt on start including overshoot (RTFQ!)
- Loud pops/bangs...rpm down EGT up = comp stall
- The over voltage unit takes the generator offline at 30.2 volts
- The biggest user of battery juice is the pitot heat
- Igniters powered by 28vdc ess serv bus
- something about thrust loss with the bleed valve stuck open
- No NWS with emergency gear extension
- Emergency brake/flap accumulator gauge shows nitrogen pressure
- A slow loss of HYD 2 pressure will not lose general services
- T handle separates engine from aircraft fuel system
- ATS mechanically drives N2 compressor
- 52 +/- 2 idle rpm closed bleed valve
- know engine fire indications (light, tone, etc)
- L Bar is mechanically raised and hydraulically lowered
- Skid advisory light = both stitches on and gear down
- know what the hyd gauges in the cockpit should read in start (1=3000, 2=0, brake=3000)
- single AC inverter failure = loss of ac non essential services, AC inv light, tone
- first indication of a fire on takeoff is the fire light (true)
- if the fire light comes on then turns off = temp below 300 or burn through
- EGT/RPM light = 650 +/-8 or 112.4 +/-1 N1 RPM
- Know what triggers a NWS caution light
- N2 Accessory gearbox = oil pump, hyd EDP's, LP Pump, DC Generator, etc
- no emergency flaps with total electrical failure
- the gts powers the ats on start
- loss of both inverters = loss of AC services, light, tone (all the above I think)
- Generator failure = loss of hud, right mfd, gen warn light, master caution
- NO VERIFICATION of INTERNAL tailpipe overtemp
- accumulator preload charges (Hyd flight control = 1100 +/-50, 1300 +/-50 for brake)
- 5 x to turn off anti skid switch (after hyd 1 loss, after shurdown prior to setting pk bk, boat, skid caution light, blown tire)
- 300 amp fuse between generator and gen bus
- hook extended by gravity assisted by snubber
- F Press light for boos pump failure
- ECA failure dual lane full and no trim indicaitons

- Flameout below 1500 AGL AND 180 kias = eject
- GINA made up of gps and ins
- HOT START INDICATIONS (there are 2 answers so RTFQ)
- how many BIT displays are there? 2.. Weight on Wheels and Weight off wheels
- Alignment of gina complete when waterline replaces by velocity vector
- Mode C = altitude reporting for transponder
- extending the flaps requires 28vdc power
- yaw dampener trims rudder via ARI
- IFF DGRD in bit page: fail, standby, off
- If CAUG light illuminates in flight, lose ARI, SBI, rudder trim, yaw dampener
- WILL have tacan with generator failure
- IFF is only CNI device with 2 antennas
- From Mr Carrier himself: "To enhance the efficiency of the ECS heat exchangers, inducers powered by bleed air by the IPRSOV which is commanded open by: Gear down and ECS selected to any position other than off"
- Ram air in cockpit contriled by a Weight on wheeles switch ECS air flow knob
- rudder shakers and tone and best indications of an impending stall
- DEU failure limits you to a GCA
- Oxygen contractor filters litrogen and other contaminants
- hung start indications (rpm stuck at 45 and egt rising)
- rapid roll during flap/slat transition = split flap/slap
- ILS needles displayed when ILS selected on HIS
- Dislay system gets date from gina and pitot static system
- full deflection of CDI in tacan, VOR, wpt steering (all the above I think)
- OPGO on bit page = single channel
- engine switch on in both cockpits energizes starter control unit
- raising the finger lift will: allow throttle to go below appr idle, go past fight idle to off, prevent inadvertent pull to off in a rapid retraction
- display switch to ORIDE to keep L MFD on during a gen failure
- electronic sequencer determines ejection mode
- CANOPY light whenever canopy unlocked....get a tone of unlocked and go past 95% rpm.

Timms Gouge EP Test 1, old gouge still holds as well

1. Engine fails to light off with in 15 sec from putting throttle to idle – Wet start
2. Audible bang, Increase EGT, RPM lowers = Engine Stall
3. During descent passing 7000 MSL, ACCEL Caution light – expect slower engine acceleration during an approach
4. HYD 2 pressure failure, RAT will maintain HYD 2 pressure at 2500-3000
5. Taxi clear of active, NWS malfunction- aircraft would be steered by differential braking
6. If aborting with a blown tire, set ANTI-SKID switch ON – (False)
7. ECA 2 Caution light- land as soon as practical and do not exceed 90% above 20K MSL, 95% below 20K MSL, do not exceed 600 deg EGT
8. Just flamed out, airspeed is 150 and at 1000' you should – Eject
9. Abort procedures- IDLE, BOARDS retract, BRAKES, HOOK (1000 ft. prior)
10. LP PUMP light in flight- avoid abrupt throttle movements and set min for flight
11. What requires an immediate shutdown on deck w/out 2ndary indications- FIRE light illums
12. When flying @ 15K, TP HOT light- Throttle idle (min safe flight)
13. On cat stroke, aircraft settles- Throttle full, check 24 AOA, if settle continues Eject
14. Launch bar system is – electrically controlled, mechanically extended, and hydraulically retracted
15. Engine failure on cat stroke and unable to eject- ditch straight ahead
16. For HYD 1 or HYD 2 Failure 1st consideration- Reduce A/S below 300 KIAS/.6 MACH
17. After F PRESS light illuminates and checking engine switch on you should- avoid negative G maneuvers
18. While cruising 23K you get FIRE light, with 2ndaries- EJECT (not prepare for controlled ejection)
19. NWS AUG system provides yaw rate feedback in high gain nose wheel steering – false (yeah we had no f'ng idea either)
20. 3-4l questions that deal with procedures for COMP Stall and EGT RPM Light. Know the first 3 steps in that procedure and that's 3-4 free questions.
21. If you get a HYD 2 failure, monitor HYD 1 Presssure
22. Executing the Engine Fire Emergency Shutdown on the Ground, state the procedures verbatim

(A,E,B,D,F, if different use PCL!) 1) Throttle OFF, 2) Engine Switch-OFF (both Cockpits), 3) Fuel Shutoff – PULL, 4) BATT switches – OFF, 5) Emergency Egress

23. During ACM you get a HYD caution light, RAT caution light and then the HYD light extinguishes you should: Reset HYD 2 (correct but not the first thing to do, Instructor took note & future test may be changed)

24. First step in the EGT/RPM warning light: Neutral/IDLE

25. If NWS AUG goes on during taxi, you should: Momentarily press NWS button.

26. GTS Fire on deck: Emergency Shutdown

EP Test 2

1. Descend below 10,000 feet cabin alt. with an OBOGS Failure
2. Eject if below 1500 AGL and below 180 knots
3. Electrical fire immediate action- GEN switch off
4. (2 questions) If put you get a SLATS caution light- maintain a/s less than 200 KIAS
5. Smoke and fumes in cockpit- Altitude descend below 25K, AIR FLOW knob off
6. With directional control in question, short field arrestment: hook down, land 50 ft prior
7. Uncommanded roll or yaw when transitioning flaps/stats- return flaps to previous position
8. OXYGEN warning light, system won't reset set OBOGS/ANTI-G switch off
9. Blown tire after touchdown- Rudder (counter swerve) and Go around if able
10. GINA failure- all the above
11. Abnormal MFD- blinking screen, invalid display, blank screen, stuck push button
12. With barometric altitude failure- Use radar alt if below 5K AGL, use cabin altimeter if unpressurized, and check PITOT Heat is on
13. Engine failure catapult- Eject
14. Immediate Airstart- Pull Emerg. Oxygen actuator, throttle off, GTS button, idle
15. C AUG light on lose- ARI, SBI, yaw dampner, and rudder trim
16. GTS fire- Emergency engine shutdown
17. Cabin pressurization failure- descend below 25K MSL
18. Compressor stall- IDLE, Neutralize, monitor
19. Blown tire on T/O roll and abort- rudder (counter swerve), Anti-skid off
20. Blown tire, no arresting gear- flared, half flap
21. CANOPY light in flight- Airspeed below 200 knots
22. Multiple MFDs f'd up- momentarily set Display Power switch to RESET
23. Total Electrical failure can't restore power- No flap approach and make an arrested landing
24. Windmill airstart parameters- below 25K, 13%, min 250 kts
25. Abort takeoff- Idle, extend, brakes as req, hook 1000 ft prior
26. Electrical failure- no load shedding (eliminates 3 out 4 answers)

27. Pitot static failure question- standby VSI
28. Settling off catapult- throttle check full forward, 24 AOA
29. Launch bar warning light after cat launch- divert to shore, min. descent rate landing with arresting wires removed
30. Gen failure- confirm engine operation, reset Gen switch
31. Wheel brake failure go around not possible- brakes release, anti skid- off, attempt braking, hook down 1000 feet prior
32. AC INV light- reset number 1 and 2 inverters
33. Hook light still illum after hook should be down- hook is not fully extended
34. Anti-skid failure- release brakes, anti-skid off, brakes as required
35. 28 VDC gen bus failure will not remove- primary stabilator trim
36. Fire light in flight- Idle, instruments check, secondary-eject

Metro Gouge

- 1) southern Oregon is clear
- 2) Cold front is blowing from the northwest
- 3) The stronger side/more turbulent side of the jet stream is the northern side
- 4) The jet stream dips south in the winter, is higher in altitude, and stronger/faster (check this for accuracy)
- 5) Turbulence is experienced most when (Mike has the controls)
- 6) wing tip vortices begin on takeoff/rotation for larger aircraft
- 7) A low pressure gradient is strongest/lowest pressure is where the isogonic lines are closest together... location E in the figure is the best answer.
- 8) ATIS information is obtained from the UHF/VHF radio... not the VOR voice
- 9) It is not always against OPNAV-3710 to go into a SIGMET. However the Area 2 in that particular question is the best answer. Read the whole question if need be, but area 2 (southern Texas I think) is the best answer
- 10) 1014.2 millibars (or something like that is one of the answers, everything else is attractive, but it's the only one that makes sense)
- 11) when flight planning, use actual weather where you are, actual and forecasted enroute, and and forecasted at your destination
- 12) Know how to read a METAR. A METAR is a report of what is.
- 13) Know how to read a TAF and know that something BECOMING and a TEMPO can both be at the same time and should both be referred to for weather minimums
- 14) 200-4 are mins for a single piloted aircraft (departing but also generally)
- 15) Look at the forecast for TKS and figure out the lowest mins: 500-1 (it's plain as day, but I looked at the stupid question for like 5 minutes before I figured out what it was asking, poorly worded)
- 16) An AIRMET is the same as a SIGMET but it is published for less serious reasons.
- 17) A DD-175-1 must be filed for all IFR flights in IMC conditions... IFR is the answer, not all flights
- 18) In one question they ask, the weather brief is good until? The answer is next to the void box at the bottom of the WX brief, it has a void time of 1630, a rebrief time of 1710 and a new expiration next to that (1750/1730) that's the right answer. Look at the extension time.
- 19) When should a PIREP be made? All of the above
- 20) If caught in a microburst, add full power and wave off/abort (basically do what Nate does on 75% of his landing attempts)
- 21) The T-45 is considered a single piloted aircraft! Disregard the fact that we have two radios
- 22) In one question about forecasted turbulence, its moderate to severe #3 (like Ted's gambling addiction)
- 23) Severe Turbulence over Maine is from the surface to 12,000
- 24) The freezing level in North eastern Oregon is 10,000 MSL (Interpolate) Low Level Significant weather Prognostic chart
- 25) Weather over Northeast: Light Snow and tops of 14000
- 26) When planning a cross country from Florida to Mississippi (or something like that) the wind is a left crosswind at 65 kts

North side
wind strongest
fastest

Low
clouds
low visibility
warm front
Wanna go
replay w/

27) Ceilings are always in AGL

28) 2 winds aloft questions. Know how to read them when above 100 knots

29) Know what a stationary front looks like

30) On a weather depiction chart there will be a question about whether an area is VFR, MVFR, or IFR. Its VFR despite the $\frac{3}{4}$ cloud cover symbol

31) Do not go through a Thunderstorm. Divert (unless you were one of 1600 to get into Notre Dame)

32) If you do end up going through a thunderstorm, set attitude and ride it out. Don't chase needles

33) Thunderstorms threaten aircraft with lightning, icing, hail

34) A warm front has stratiform clouds and rain on either side. (may as well know what the different fronts do, but the question on the test is a warm front)

35) One question is reading a METAR and asks which answer is right... the answer is the visibility is increasing at the station (they put the METARs from latest to earliest to confuse you so it looks like vis. goes down but look at the times... it (visibility) really increases)

36) One question says you are going through CA and Oregon West of the Cascades on a flight... they ask you what the conditions will be. The answer is both: the conditions before and after 1700z. make sure you look it over to doublecheck

37) Know about structural icing- decreases lift because it changes the shape of the airfoil

38) On a radar summary chart they ask about what's going on in Maine and into the Atlantic... the answer is snow observed and echo tops at 14,000' MSL

39) One question asks about San Francisco (SFO) the answer is that the weak high pressure system will be formed in coastal sections by 0200Z (something close to that)

40) They ask about a forecast for Texas and Oklahoma... answer is something to the effect of occasional severe turbulence below 6,000'

41) One asks about filing a weather brief, answer is 2 1/2 hours prior to take off and 30 min after, its void

6920 689 512 210

22 400Z

Learn Fronts

Flight orig in OR and moving

answer c. both A+B

Core P w/ alt
wind P

Jetstream
Shifts
South
Winter

INAV Timms Gouge- If you can't pass you're going E2s

Cleared for published DP on IFR flight plan. After takeoff "climb and maintain FL350..."

- Climb immediately to FL350 and proceed via direct flight to the DP transition point.

On a flight from NAS Meridian to Montgomery Regional, prior to commencing a HI ILS aprch, tune and ID NAVAIDS. Which TACAN should be tuned and IDed?

- Channel 97 (MXF)

When flight planning, an airport designated "Official Business only"

- Be designated as an alternate on an IFR flight plan

For all flights in a jet, 3710 requires fuel from takeoff to landing plus a reserve of:

- 10%, but not less than that required for 20 minutes flight at max endurance for 10,000 MSL

Which of the following are true concerning HI-TACAN RWY 19L approach chart to NAS Meridian

- You must maintain 3000' between the 328 and 358 degree radials

With all NAVAIDS and facilities operating, takeoff mins for a standard instrument rated pilot departing RWY 9 at NAS Jax would be

- 200 ½

When on an IFR flight plan and cleared to an intermediate point not in a filed route, short of filed destination, should not accept the clearance without first obtaining

- an expected further routing

On a MA at NAS Meridian you decide to go to your alternate, Maxwell AFB. CAT-C mins for an ILS RWY 15 circle to land RWY 33 approach will be

- 500 1½

You are in GCA box pattern, within 25NM of the filed in IMC, when comms are lost. From here, intercept a published instrument aprch maintaining ___ until established on a segment of aprch.

- the highest of your last assigned or published MSA

Your destination is forecasting 800 1½ Published aprch procedures (NW winds) at alternate, Columbus AFB, lowest forecast wx conditions for filing would be ___. ILS and ALS inop

- 700 2

To be cleared for a visual aprch, must report to aprch control field or interval in sight and wx must be reported as

- a minimum of 1000-3

Inadvertant IMC and lose sight of the runway while circle to land, you should:

- Execute the MA for the original aprch procedure

Min forecast wx needed to plan for NAS Meridian (RWY 1L) as you destination? Maxwell AFB is alternate and forecasting 900/3. NAS Meridian NOTAMS: GCA O/S ALL, ALS O/S RWY 1L

- 500 1½

A LOC transmitter has range of ___ NM within 10* of centerline and ___ NM within 35* centerline.

- 18, 10

Reading back an ATC clearance implies acceptance of the clearance, unless ___.

- you state refusal and reasons

Executing the ILS RWY15 aprch to Maxwell AFB, you should be at ___ AGL when crossing rwy threshold

- 50'

The following additional reports should be made to ATC or FSS facilities without a specific ATC request (when not in "radar contact:")

- All of the above

ID the true statement concerning DPs and/or STARs

- DPs are designed to provide transition from the airfield to the appropriate enroute structure, and their use on the DD-175 is encouraged but is not mandatory.

Your MDA for Maxwell AFB TACAN RWY 15 aprch (with ALS) is published as ____

- 500 MSL

Takeoff is allowed if the current weather at the departure airport is at least ____ and meets or exceeds the published precision mins for the runway in use

- 200 ½

While on the Maxwell AFB TACAN RWY 15 aprch, when would you execute a MA if you did not have runway in sight?

- At 1.5 DME from the TACAN

Flying on J2 btw Crestview and Seminole you would change your TACAN to CH 122 when you are at the CEW 088/

- 80 DME

Select the choice that best describes the service volume characteristics of an (L) VOR navigational system. Range ____

- up to and including 18000 AGL out to 40 NM

At Maxwell AFB, the PAPI system availability will be

- denoted in the IFR Sup by a "50" under lighting

The star next to the ATIS found in the upper left corner of the TACAN aprchs to Maxwell AFB denotes that:

- the ATIS operates on a noncontinuous basis

Of the following situations, the one which does not require a voice report (NOT RVSM) is:

- reaching a newly assigned altitude

You request a PAR aprch to RWY 19L at NAS Meridian. Your DA in a single-piloted aircraft for an aprch to that runway would be:

- 517 MSL

The theory and operating characteristics of a GPS system are

- GPS uses passive receivers and triangulation computations from signals received from at least 4 satellites to determine lat, long, and vertical position

Enroute to NAS Meridian, you listen to Meridian ATIS, which gives airport info, but no wx info. You know that Meridian is ____

- VFR, with a ceiling of at least 6000' and 5 miles vis or better

Planning for a flight from NAS Jax to Birmingham...for filing purposes, based on the above info, what is the lowest weather mins required at Columbus?

- 400 1

When planning for a direct flight btw two high alt NAVAIDs in the jet route structure, do not exceed the normal anticipated interference free reception range of each NAVAID, which is ____
- 130 NM

Planning for a flight from NAS Jax to BAM ETA 1500Z and alt of Columbus AFB

Your filed destination is NMM with arrived weather best to be 300-1. The alt field has PAR and TCN. The approach min weather required at the alt would be TCN min plus 300-1

If you experience radio failure enroute in IMC, you should maintain...the last assigned alt but no lower than the min enroute alt for each route segment.

You are halfway enroute filed destination on extended IFR flight at FL 290. Radio fails, in VMC and dest has marginal weather. If you determine you can maintain VMC in flt...squawk 7600 maintain VMC and something traffic pattern and land at a suitable airport.

Which publication would provide information on preferred IFR routing...FLIP AP/1

You are cleared to the PIGMY IAF @ NMM with EFC 1610Z. Presently at or above 14K. enroute you experience radio failure so as to depart PIGMY at 1610Z or after descend and commence approach.

30 miles SE of MGM at FL350 you are directed to change freq. unable on assigned freq attempt to contact center on...319.1 CATC center alt subdivided.

For a canceled DP to be reinstated ATC must...indicate the portion of the DP routing that still applies and reinstated altitude restrictions.

You have been cleared via DP. After takeoff, control advises climb and maintain FL350...

PAR pickup question last comm. Pass through highest initiated penetration alt of the IAFs at airfield.

High alt NAVAID 130

All hail the glory that is Timms BI test

PPH during S1 climb: 1500

Config on PAR downwind: 200 kts, clean

Obtain field altimeter setting during penetration checklist

Aural ID (morse code) during penetration check determines reliable navigation information and: verify the correct station is tuned

HI TACAN/VOR DME approach uses 4000 to 6000 FPM rate of descent, and no more than 30° AOB

Intercept an arc from a radial, 1% of ground speed

FAC for a surveillance approach must give you: missed approach instructions, position, location of MAP. NOT time check

Barometric altitude info comes from: A,E,H

Complete pitot static system failure, you lose mach and airspeed indications, barometric altitude. NOT vertical velocity

PAR approach, you are given precise glideslope, range, and heading information (True)

S3 pattern is the S1 pattern with ½ SRT and reversal at beginning of 2nd descent

Loop commenced at 380 kias, and 96% rpm

Missed approach is required: all of the above

VHF navigation system failure with VOR or ILS steering selected:
VOR/ILS indications removed

Using 30-degree method, intercept for an inbound course of 300°: 220°

15° heading change requires 15° AOB max

700 PPH for S3 descent

T45C INS failure: Both of the above

During ASR, arrive at MDA before the MAP

Nose-low UA partial panel, use stby airspeed and stby attitude indicator

Intercept radial from an arc scenario: Four

How long will it take a $\frac{1}{2}$ SRT turn to turn 210*? 140 seconds

ADI fails in IFR conditions- establish a reference on the stby attitude indicator

140 kias SRT will be- 3* per second

Leveling off from penetration descent:

1000' above level off altitude decrease pitch by $\frac{1}{2}$ and retract speed brakes, then use normal lead point

Nose-high UA partial panel- stby attitude and AOA indicator

Each dot on the CDI with TACAN boxed is: 5* deviation

BI nose high UA recovery begins with: MRT, speed brakes in, stick pressure to obtain 5-10 units AOA

If AOA indications fail: use airspeed indicator

VOR penetration, begin level off at 10% of vertical velocity

Dirty stall, maintain 23-24 units AOA

Will lose 6000' performing Split S

Transition from climb to descent in S1 3 seconds or 100 feet prior

Total GINA failure indications: blanking of attitude and heading info on ADI, HSI, and HUD

Turn pattern, 30* AOB for 60* heading change

Nose-low UA, speed brakes extended if: airspeed is approaching or exceeds 300 kias

Three steering options are: WYPT, TCN, and VOR

45 degree intercept question: 135*

Ground speed calculation: 330 kts

During final approach on PAR, must communicate every 5 seconds

DA SHIZNIT
AERO GOUGE (JAN '05) This is the test -- know exactly these!

29-30 Units

26 Units

17 Units

Mach .92-.95

3,450'

Rudder shakers

161 CAS

15,000

Airspeed 50-120

Up & to right addition
~~Down/left w/weight reduction~~

TEMP, ALT, RPM

Sweep and Low thickness-to-chord ratio

Longer roll out on landing

Forward increasing stability

Decrease stability, Increase maneuverability

1 AOA all configs

Strong Directional, Weak Lateral

Mach Tuck under

✓ 125 (not 129)

✓ LOW

✓ Fuel burn & Gear + flap movement

~~Fuel burn does not move CG~~

Force Divergence Mach number

NOT IN
BACK BUT RIGHT
FOR TEST

High AOA stall decrease C_L

(Component of lift)
Coefficient

TAS/SOS

Max. Endurance, Max. Climb Angle(V_x)

Approach stall occurs at what AOA?

Clean stall occurs at what AOA?

Optimum AOA NO FLAP(Any flap)?

Some wing heaviness may be experienced, up to 3/4 lateral stick may be required to maintain wings level flight.

T/O w/full flap DRY Runway

Initiated by front left, but shakes pedals in both cockpits

BINGO

45 upright spin ROD is ?

AS for -60 inverted spin

Tr Curve shift _____ w/weight addition.
available

Thrust is effected by ... (not AS)

Improve transonic flight (not leading edge radius)

Density Alt increases, IAS stays constant, TAS increases

Two pilots move CG....

CG aft limits

External stores decrease AOA by

Spiral Divergence

Aerodynamic center moves aft, nose pitches down, .8

Max Abort

If density ALT is high, PA is high so density of air is...

~~Upper fuselage/Center (fuel tanks)~~ which shift CG.

The Mach number where the coefficient-of-drag and total drag increase rapidly due to wave drag.

Disadvantage of swept wing

Mach =

The Airspeed/AOA for _____, _____, & _____ found at L/D

& Max. Power off glide RANGE

Neutral

Area Rule

Reduce "Rudder Buzz"

Vortex Generators

✓ Stall Strips

✓ >250 lbs of pressure

Know spin recovery for upright and inverted

Nose pitch up 20 degrees

Uncontrollable pitch up

✓ Q

✓ Critical Mach Number

Dutch Roll

6.5, 3.0

Nose away from X-wind

✓ 21.5 units

Upright and Inverted spins

0 AOA, Neg. G's, Turn needle
direction of spin

Moving forward of CG question; If CG moves forward what happens, answer says maneuvering and then stability.

0.8

max.

Trust is _____ on longitudinal stability

NOT a T-45 Design

Rudder shock stabilization strips

Improve aerodynamic stability, performance and /or flight control effectiveness. Force wing downwash further below the stab, eliminating pitchup tendencies. Also improve slow speed flight by redirecting wing airflow and reenergizing the boundary layer.

Cause the root to stall at lower AOA by inducing premature boundary layer separation at high AOA. The wing tip is more prone to stall because of lower camber and shorter chord length at the tip. Boundary layer is also lower at the tip, aggravated by spanwise flow and leading edge vortexing.

Force req'd to overcome Rudder blow out

Accelerated stall w/no external stores, second indication of impending stall

AC moves in front of CG at stall. CG in front of AC = More stability less control. CG close to AC = less stability more control.

Air mass flow in slugs per second

The free-stream Mach number at which a local Mach number of 1.0 is attained at any point on the body under consideration

Low Airspeed, Weak directional, Strong lateral

G limit question

Cross wind does what to nose of T-45

Stall warning at 1/2 flaps

Rudder opposite for what type of spins?

Inverted spin is characterized by?

Critical mach #, slight nose down pitch

NOT IN BOOK,
IS ON TEST

USE
CHART!!
in PCL!

AERO GOUGE

1. Approach Stall AOA? 29-30 / OKIAS
2. Clean stall AOA? 26 / OKIAS
3. Optimum AOA NO FLAP? 17
4. $\frac{3}{4}$ Stick w/heavy wing Mach .92-.95
5. 3,450' t/o w/full Flap dry RNWY.
6. Rudder shakers both c/p both pedals
7. Bingo 161 CAS
8. 45° upright spin 15,000 ROD
9. Airspeed 50-120 -60° inverted spin → 50 - 120
10. Tr curve Down/left w/weight reduction
11. Thrust effected by Temp, ALT, RPM ~~Not Airspeed~~ RAT
12. Sweep and Thickness to cord Ratio Improve transonic flight (not leading edge radius)
13. Density ALT increases IAS stays constant TAS increases: longer Rollout on TAI landing
14. Two pilots CG moves forward increasing stability
15. CG forward limits increase stability
16. CG Aft limits decreased stability
17. External stores decreases stall by 1 AOA all configurations
18. Spiral divergence- Strong Directional, weak lateral. • strong direct forward
• weak lateral.
19. Mach tuck under- Aerodynamic center moves Aft, nose pitches down
20. Max abort 125 (not 129)

21. ~~Upper fuselage/center (fuel tanks) fuel burn do not move~~

22. Force Divergent Mach Number- Know what it means-(P 15) *.85 THE MACH # W/ C_D AND T_D ↑ DUE TO WAVE DRAG*

23. Disadvantage of swept wing- High AOA stall decrease C (component of lift)

24. Mach = TAS/SOS

25. L/D max > max endurance, power off glide, climb angle

26. Thrust is Neutral on longitudinal stability

27. Area Rule Not a T-45 Design

28. Rudder shock stabilization strips- Reduce "rudder buzz"

29. Vortex Generators- 1) Prevents wing downwash from interfering with stabilator
2.) slow A/S performance increases, lowers stall speed
3.) maintain boundary layer pg 23, 31

30. Without boundary layer separation bullshit a swept wing stalls tip first Pg 29 stall strips

31. Rudder blowout >250 lbs of pressure

32. OCF Rudder, Lateral input control, opposite for INV and Upright, upright Aileron same, inv cross control

33. Good luck on the other 7

1034 34. Accelerated stall w/no external stores- pitch up 20 *AOA/degrees* second indicator of impending stall pg 33

B51 35. Thrust effected by RPM, ALT, TEMP

B45 ? 36. AC moves Front of CG at stall- Uncontrollable pitch up

B55 37. Q: is mass Air flow in slugs/ Sec

38. Critical Mach # Definition- Free stream mach # @ which a local mach # of 1.0 is attained at any point on the body under consideration, .8 for the T-45

B 47 39. Dutch Roll- low airspeed, strong lateral, weak directional

40. G limit question- ~~0.8-2.0~~
41. Crosswind does what to Nose- nose away from X-wind
42. Stall at 1/2 flaps- 21.5 units warning at
43. Rudder opposite turn needle for both upright and inverted spins
44. inverted spins characterized by- 0 AOA, neg G's, Turn needle direction of Spin
45. Aileron positions reversed for spin recovery (upright + Inverted) upright into turn needle, Inv away from needle
46. T-45 turns away from crosswind pg48

	<u>Warn</u>	<u>Buffet</u>	<u>Stall</u>
Clean	21.5/ 10IAS	25/1-2	26/0
Approach dirty	21.5/10IAS	28/ 1	29-30/0

Moving forward of CG question, If CG moves forward what happens, answer says maneuvering and then stability

Upright spin

	<u>A/S</u>	<u>Descent</u>
+30	INC 180	12,000 fpm
+45	stdy 100-110	15,000 fpm

INVERTED Spin

-60	50-120	15-20,000'
-40	100-160	15-20,000'
-25	140-200	11,000'

AERO GOUGE

1. Approach Stall AOA? 29-30 UNITS
2. Clean stall AOA? 26 UNITS
3. Optimum AOA NO FLAP? 17 UNITS
4. $\frac{3}{4}$ Stick w/heavy wing Mach .92-.95
5. 3,450' t/o w/full Flap dry RNWY
6. Rudder shakers both c/p both pedals, INITIATED BY FRONT LEFT
7. Bingo 161 CAS
8. 45 upright spin 15,000 ROD
9. Airspeed 50-120 -60 inverted spin
- ✓ 10. Tr curve Down/left w/weight reduction
- ✓ 11. Thrust effected by Temp, ALT, RPM Not Airspeed
12. Sweep and Thickness to cord Ratio Improve transonic flight (not leading edge radius)
13. Density ALT increases IAS stays constant TAS increases: longer Rollout on landing
14. Two pilots CG moves forward increasing stability
15. CG forward limits increase stability
16. CG Aft limits decreased stability
17. External stores decreases stall by 1 AOA all configurations
18. Spiral divergence- Strong Directional, weak lateral
- 24 19. Mach tuck under Aerodynamic center moves Aft, nose pitches down
20. Max abort 125 (not 129)

3450

161 CAS 1130

125 Max abort

DENSITY- ALTITUDE IS HIGH THEN PA IS HIGH SO

Fuselage (Upper) do not move CG

Speed →

- 21. Upper fuselage/center (fuel tanks) fuel burn do not move CG
- 22. Force Divergent Mach Number- Know what it means-(P 15)
5% > Mach # where there is a rapid ↑ in C_D due to formation of normal wave Mach #, where coefficient of drag T_D increase
- 23. Disadvantage of swept wing- High AOA stall decrease C_L (component of lift)
due to wave drag

24. Mach = TAS/SOS

25. L/D max > max endurance, power off glide, climb angle *side ratio*

26. Thrust is Neutral on longitudinal stability *Two*

27. Area Rule Not a T-45 Design

28. Rudder shock stabilization strips- Reduce "rudder buzz"
improve stab, perfor, flight control effectiveness

29. Vortex Generators- 1) Prevents wing downwash from interfering with stabilator performance
2.) slow A/S performance increases; lowers stall speed *adversely in the transonic flight regime*
3.) maintain boundary layer pg 23, 31 *force downwash below stab*

30. Without boundary layer separation bullshit a swept wing stalls tip first Pg 29 stall strips
stall strip *Spin* *Upng OProd* *Down OProd*

31. Rudder blowout >250 lbs of pressure

32. OCF Rudder, Lateral input control, opposite for INV and Upright. upright Aileron same, inv cross control.
Upright opp rudder same aileron
inv opp rudder opp aileron

~~33. Good luck on the other~~

34. Accelerated stall w/no external stores- pitch up 20 second indicator of (buckling) impending stall pg 33 *and roll to either side*

35. Thrust effected by RPM, ALT, TEMP, RAT
NOT AIS

Very Slow →

36. AC moves Front of CG at stall- Uncontrollable pitch up p.45
Acft Center

37. Q: is mass Air flow in slugs/ Sec

38. Critical Mach # Definition- Free stream mach # @ which a local mach # of 1.0 is attained at any point on the body under consideration, .8 for the T-45

39. Dutch Roll- low airspeed, strong lateral, weak directional
<220 KIAS

USE CHART
IN PCL!!
<5000 MSL

40. G limit question- ~~2000~~
41. Crosswind does what to Nose- nose away from X-wind
42. Stall at 1/2 flaps- 21.5 units warning at ~~10~~ not man
43. Rudder opposite turn needle for both upright and inverted spins
44. inverted spins characterized by- 0 AOA, ~~neg G's~~ ^{-1.1g}, Turn needle direction of Spin
45. Aileron positions reversed for spin recovery (upright + Inverted) upright into turn needle, Inv away from needle
46. T-45 turns away from crosswind pg48

	<u>Warn</u>	<u>Buffet</u>	<u>Stall</u>
Clean	21.5/ 10IAS	25/1-2	26/0
Approach dirty	21.5/10IAS	28/ 1	29-30/0

UP → Same
21.5 25 26
21.5 28 29-30

Moving forward of CG question, If CG moves forward what happens, answer says maneuvering and then stability

↳ more. ↳ less

21.5 25/12 26
21.5 28/1 29-30

<u>Upright spin</u>			
A/S	Descent	AOA	
+30 INC 180	— 12,000 fpm	28	
+45 stdy 100-110	— 15,000 fpm	30	

+30 — 180 — 12000
+45 — 100-110 15000

<u>INVERTED Spin</u>		
-60	50-120	15-20,000'
-45	100-160	15-20,000'
-25	140-200	11,000'

50-120 15-20000
100-160 15-20000
140-200 11000

UP < 30 (28) 180 — 12000
45 (30) 100-110 — 15000

(200) Down < -60 50-120 — 180-20
-45 100-160 15-20
-25 140-200 11

✓ Disadvantage of Swept wing - Stall at High AOA and decrease Coefficient of Lift

32

✓ L/D Max = Max Endurance, Max Climb Angle, Power Off Glide

Thrust is Neutral on Longitudinal Stability

✓ Area Rule not part of T-45 Design

p 28

✓ Rudder Shock Stabilization Strips - Reduce "Rudder Buzz"

✓ Vortex Generators

Prevents wing downwash from interfering with stabilator
Slow Airspeed performance increases - lowers stall speed
Maintain Boundary Layer

✓ Without Boundary Layer Separation Aids - Swept wing stalls tip first

✓ Rudder Blowout - Requires >250 lbs of pressure to counter

✓ Accelerated stalls with no external stores - 20° Pitch Up is second indicator of impending stall

Thrust affected by - RPM, ALT, and TEMP

✓ AC moves forward of CG at stall - Uncontrollable pitch up - Similar answer for this question
OPPS I got it wrong

✓ Mass Airflow in Slugs/Sec = $\dot{Q}$

✓ Critical Mach Number - Free stream Mach # at which a local Mach # of 1.0 is attained at any point on the body under consideration - .8 for the T-45

✓ Dutch Roll - low airspeed, strong lateral, weak directional

✓ G Limit Question - ~~6532000~~ USE CHART IN PCL

✓ Crosswind - Pushes Nose Away from Wind

↑ *transitions* →

TRUE AOA	INVERTED SPIN MODES (true AOA)		
	-60 degrees	-40 degrees	-25 degrees
Indicated AOA	0	0	0
Airspeed	50-120	100-160	140-200
Altitude	Decreasing rapidly	Decreasing rapidly	Decreasing rapidly
VSI	6,000 fpm descent	6,000 fpm descent	6,000 fpm descent
Turn needle	In direction of spin	In direction of spin	In direction of spin
Descent rate	15,000-20,000 fpm	15,000-20,000 fpm	11,000 fpm
Nose attitude	30 degrees down	50 degrees down	65 degrees down
g's	-1 to -1.4 g's	-1 to -1.4 g's	-1 to -1.4 g's

- ✓ Rudder Full Opposite Turn Needle for both upright and inverted spins
- ✓ Aileron Full with Turn Needle for Upright Spins
- ✓ Aileron Full Opposite Turn Needle for Inverted Spins

May need 3/4 Lateral Stick to maintain wings level during heavy wing between M .92-.95
 At M .95 Roll Rate 27° with Full Stick
 Above M .95 Heavy Wing goes away

- ✓ Rudder shaker system shakes both pedals in both cockpits

T_R Curve - Down/Left with weight reduction

- ✓ T_A is affected by Temp, Alt, and RPM - not Airspeed p 51

Sweep and Thickness to Cord Ratio Improve Transonic Flight - Not leading edge radius

Density ALT increases - IAS stays constant, TAS increases = Longer rollout on landing

Two Pilots - moves CG forward increasing stability

- ✓ CG - Forward increases stability - Aft decreases stability

Upper Fuselage and Center Fuel Tanks - do not move CG during fuel burn

- ✓ Spiral Divergence - Strong Directional, Weak Lateral

Mach Tuck Under - Aero Center moves Aft - Pitches Nose Down

Max Abort 125 - not 129

Force Divergent Mach Number p 16

Approximately 5% critical Mach - rapid rise in drag coefficient due to the formation of the normal compression wave.

- ✓ Mach = TAS/SOS

17 knots = 60 ft/sec
 Alt 1700

Alt 1700 = 114
 12 119
 13 124
 14 128
 15 133

- ✓ Approach Stall AOA
29-30 units
- ✓ Clean Stall AOA
26 units
- ✓ Optimum AOA - No Flap
17 units
- ✓ External Stores

Decrease stall AOA by 1 unit for all configurations

	ARTIFICIAL STALL MARKING	STALL EFFECT	STALL
CLEAN	21.5 UNITS / 10 KIAS 21.5 / 10	25 UNITS / 12 KIAS 25 / 12	26 UNITS / 10 KIAS 26
APPROACH	21.5 UNITS / 10 KIAS 21.5 / 10	26 UNITS / 11 KIAS 26 / 11	29-30 UNITS / 10 KIAS 29-30

Due to stall

TRUE AOA	UPRIGHT SPIN MODES (true AOA) +30 degrees	+45 degrees
Indicated AOA	Above 28	30
Airspeed	Increasing through 180	Steady 100-110
Altitude	Decreasing rapidly	Decreasing rapidly
YSI	> 6,000 fpm descent	> 6,000 fpm descent
Turn needle	In direction of the spin	In direction of the spin
Descent rate	12,000 fpm	15,000 fpm
g's	Positive	Positive

SPIN INDICATOR TYPE	SPIN TYPE	
	UPRIGHT	INVERTED
AOA	28-30	0
Airspeed	100-180	50-200
Altitude	Decreasing rapidly	Decreasing rapidly
YSI	> 6,000 fpm descent	> 6,000 fpm descent
Turn Needle	Full in direction of spin	Full in direction of spin
g's	positive	negative 1-1.4
Recovery AOA	decreasing past 20 units	increasing past 5 units

Significant Mach Numbers

* Maximum level flight speed	M 0.83 NOW .85
* Maximum speed reached	M 1.04
* Critical Mach Number (M_{crit})	M 0.8
* Force Divergent Mach Number (M_{fd})	M 0.85
* Slight Buffet	Passes M 0.9

Longitudinal Control

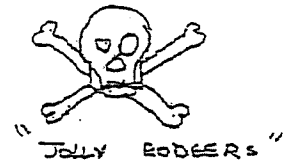
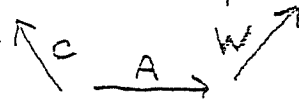
* Slight nose down pitch (Tuck Under)	M 0.8
* Predictable longitudinal control	to M 0.84
* Stick forces more sensitive	M 0.84
* Band of reduced stability	M 0.87
* Pitch oscillations	M 0.95+
* Stability increases	M 1.0+

Lateral Control

✦ Crisp lateral control	to M 0.9
✦ Roll response unpredictable	Loaded rolls
	above M 0.8
✦ Near stall roll rates	> 260° / second
✦ Heavy wing	M 0.92 to M 0.95
✦ Roll rate Degrades to 27° / sec. w/ full stick	M 0.95
* ✦ $\frac{3}{4}$ stick deflection maintain wings level	M 0.92 - M 0.95
✦ Wing heaviness disappears/Aileron authority ret	M 0.95
✦ Roll rate	120° / sec M 1.0 +

only *'s on test

M WIG DOWN UP



Approach STALL AOA? 29-30

Clean STALL AOA? 26

OPTIMUM AOA NO FLAP? 17

3/4 STICK W/ HEAVY WING

3,450' T/O. W/ FULL FLAP DRY RWY

RUDDER SHAKERS BOTH C/A BOTH PEDALS

BINGO 161 KCAS

45° SPIN 15,000 ROD

AIRSPED 50-120 -60° INV. SPIN.

T_r Curve DOWN/ LEFT W/ WEIGHT REDUCTION

TRUST_A EFFECTED BY TEMP, ALT, RPM NOT AIRSPED

SWEPT & THICKNESS TO CORD RATIO IMPROVE TRANSONIC FLIGHT.
(NOT LEADING EDGE RADIUS) !!!

DENSITY ALT ↑ IAS STAYS CONSTANT TAS ↑ : LONGER ROLLOUT LA

TWO PILOTS CG MOVES FORWARD ↑ STABILITY

CG FORWARD LIMITS DECREASED MANEUVRABILITY

CG AFT LIMITS DECREASED STABILITY

EXTERNAL STORES ↓ STALL BY 1 AOA ALL CONFIGURATIONS

SPIRAL DIVERGENCE STRONG DIRECTIONAL WEAK LATERAL

MAEW TUEK -- Aerodynamic Center moves AFT --

MAX ABORT 125 (NOT 129)

UPPER FUSELAGE / CENTER WING DO NOT MOVE CG

FORCE DIVERGENT MACH NUMBER -- KNOW WHAT IT MEANS

DISADVANTAGE OF SWEEP WING HIGH AOA STALL ↓ CL

$$MACH = \frac{TAS}{SOS}$$

L/D_{max} → MAX ENDURANCE, POWER OFF GLIDE, CLIMB ANGLE

THRUST IS NEUTRAL ON LONGITUDINAL STABILITY

AREA RULE NOT A T-45 DESIGN

TCW
A/A
63
- sizes
change's
Apert.
DISTANCE / SCANT RANGE

DEU -- Is the heart of the system

1. Controls the Mux bus (Controls Nav displays too, but that's not the "Best" answer)
2. Exchanges digital data words between itself and the ADR, GINA, and MDL via the Mux bus
3. Also has hard-wired interfaces: Analog, Video, Discrete ★
4. Is a Mission computer and Display computer, Each display has format specific options
5. On Power Up, Menu page is displayed
6. Power: 28VDC Ess, 28VDC Gen, 115 VAC N-Ess and 26VAC N-Ess BOTH 28'S VPC, VAC ESS
7. MFD Failure indicated by: Blank or Blinking display, Stuck pushbutton
8. MFD's and HUD are lost with DEU failure, you will be limited to PAR approaches (NO GYRO)
9. Display system derives flight data from GINA (pitch, roll, heading, vertical velocity) and pitot static instruments (altitude and indicated airspeed) ← NO STATIC INSTR
10. Front cockpit control of CNI will still allow rear cockpit to select any MFD display within the system.
11. MFD inputs duplicated on the DEP are LAW, CRS, HDG, Bingo

GINA *HSI FULLY FUNCTIONAL ✓ BOTH ESS + NON-ESS*

1. Receives 5 satellites, uses best 4 for GPS position
2. Composed of GPS and INA (Antenna arguably, the Kalman filter is NOT a component)
3. Alignment is complete when waterline symbol is replaced by velocity vector.
4. GINA begins realignment with auto bit initiated. Alignment is complete approximately 3 minutes after auto bit is initiated. (yes but)

HUD

1. Is an Electro-Optical device which displays flight and weapons delivery information

BITS

D GAMAR

1. Power Up, Initiated, Continuous and Manual (Manual BITS are not "system" BITS) *Radius, TAC/VOR, IFF = manual*
2. Auto Initiation of IBIT in the following order: DEU, GINA, ADR, MDL, RALT
- ★ 3. Test says "Display" BIT but means all BIT checks ↑
4. There are two BIT display pages: weight-on-wheels and weight-off-wheels
5. A single channel failure of the MUX bus will result in OPGO bit status

TACAN / ILS / VOR

- ★ 1. Know the difference between magnetic bearing and relative bearing
2. Planimetric TACAN steering will disable ILS/VOR selection (the VOR/ILS legend will not be displayed).
3. ILS localizer and GS needles will be displayed on the ADI and HUD when ILS is boxed on the HSI.
4. Full needles deflection in CDI indicates: 10 or more radial deviation in TACAN, 10 or more bearing line deviation in VOR, 4 NM or more deviation from CL in WYPT or WOS steering (gear up), .3 NM or more (gear down), or the limits of the localizer (usually 2.5 degrees) in ILS

IFF

- ★ 1. Mode test illuminates GO or NO GO lights, not ANTENNA light
2. The IFF is the only CNI system with two antennas
3. IFF DGRD on the BIT page indicates IFF power off or in STBY or an IFF BIT failure
4. Altitude squawk is Mode C

VCR

*CANNOT BOX ILS IN PLANIMETRIC
HAS 2 ANTENNAS*

1. The VCR records HUD or MFD display, ICS, radio-comms and CWS tones.

Engineering questions

- 1) GPS has how many channels? 5
- 2) Major components of the GINA are: GPS and inertial Nav assembly —and— GPS, INS & Accelerometer set (both are correct. Came from 2 diff. ?'s) D G A M E
- 3) DEU controls the MuxBus. ~~ARMED GWR~~
- 4) Ring laser gyro has few moving parts and is unaffected by g forces. It is affected by earth-rate precession.
- 5) LAMP test has no effect on fuel quantity indicator. "FUEL" light is yellow caution, not red warning light.
- 6) Rudder shakers at 21.5 units. 21.5
- 7) MFD has 3 controls, 20 push buttons and they are on the MFD bezel.
- 8) Which instrument(s) display rad alt readout? ADI and HUD
- 9) What could you use if your ADI (?) display failed? STBY AI (other MFD should default to ADI anyway)
- 10) Which instrument would verify VSI display readout failure? STBY AL, STBY airspeed, STBY ALT, STBY VSI.
- 11) If off flag displays on the AOA indicator (front cp) you can still use the AOA indexer. NO
- 12) What indication is not a failed AOA front cp indicator? Missing AOA read out MFD/HUD displays (comes directly from AOA transmitter)
- 13) INDICATIONS: Master Alert, caution tone, ~~ACCEL caution light~~, 20V on voltmeter. 21V
- What has malfunctioned? Undervoltage sensing unit.
- 14) Dual inverter failure, you will lose?— all AC services.
- 15) Engine: With normal engine instrument indications (rpm, EGT, FF) on t/o roll what other indication of loss of thrust could you have besides failure to reach line speed?
—ACCEL caution light illuminates
- 16) T or F: Illumination of the ACCEL caution light indicates the initial shot solenoid or the steam ingestion bleed valve is not in the commanded position. T
- 17) If the fire light illuminates but then goes out, what secondary indications, other than EGT/RPM light illuminating, would verify an engine fire even though the FIRE light is out?
EGT 640 C or higher and RPM is 83%
18. In a steep dive, the ECA senses an RPM increase and will send a signal to the FCU to DECREASE FF.
19. If you have an ECA failure (full trim) what changes would the engine instruments indicate?
—decrease in EGT, RPM, and FF Master alert, ~~tone~~, ECA 2 caution light

Electrical questions

20. What is lost with a single inverter failure? (p2-15) AC INV caution, tone; ~~GEN light remains~~
~~EFF~~ and you lose NON-Essential AC services.
1. With GEN functioning normally, which light(s) on CWS panel would illum if the undervoltage sensing unit failed? NONE
2. What is the min voltage of a fully charged batt? 24V

Run away trim w/ Gen Fail still means primary trim will work, standby will not.

hardware interface analog discrete video
can be mission & display computer
on power up MENU page displayed

DEU power: 28 V DC essential + g
115 AC essential

GINA - 5 channels uses best 4 for nav

HUD - electrooptical display flight & weapon delivery info

BIT - power up, initiated, continuous engaged manual

Auto initiation → DEU FWA ADR MDL RALT
on power up

AA Tacan - 63 channels/only gives slant range. 5000
A part

DEU → MDL & ADR

▷ To open air ^{Ram} (airborne) put AC switch to OFF.

▷ SBI moves stabilator and not stick

▷ MFD's: are powered by 28 VDC GEN BUS, 28 VDC ESSEB45, and 115 VAC NGR

T/F BIT PAGE comes up on power up.
→ it is the menu page

True?

▷ Anti-skid will be off if any switch is off TRUE

97% OK

what malfunction has occurred if ATC reports loss of alt. info? IFF Failure

when does the canopy light illuminate?

- anytime canopy not locked ~~when canopy is open~~

Completed ✓ list, advancing power to roll fwd for Brake check.

- master alert + PK Brake light + caution tone

→ Park Brake on + throttle beyond intermediate pos. 70%

Select high-gain NWS get master light, caution tone sounds, NWS #1 caution illuminates. The NWS system has failed.

→ whenever sys failure or nosewheel pos. doesn't match commanded position.

Tacan measures slant range between two cooperating a/c. (other answers wrong b/c the 1st "Relative Bearing", "E1 needle", etc)

1660 $\pm$ 110 psi

~~600 $\pm$ 50 psi~~ pressure at which HYD 1 (prior to AFC244) PRESS caution light illuminates

276 $\pm$ 6 cubic inches hydraulic reservoir gauge on preflight

45% N2 rpm required to use HYD 2 Reset button

42% N2 rpm below which HYD 2 bypass valve is energized open to reduce engine loads on start

10 ~~sec~~ number of full brake applications provided by brake/emergency-flap accumulator

2800 psi Max pressure provided by Hand-pump

600 $\pm$ 100 psi Press that HYD FAIL illuminates if RAT pressure drops below

Flight Controls and Trim System

20 $\pm$ 0.5° maximum rudder surface deflection

+6.6° to -15° maximum travel of stabilator leading edge due to stick movement

+3° to -8° maximum travel of stabilator leading edge due to trim movement alone

$\pm$ 15.5° maximum aileron deflection with gear down

$\pm$ 12.5° maximum aileron deflection with gear up

$\pm$ 9° maximum aileron trim deflection with gear down

$\pm$ 6° maximum aileron trim deflection with gear up

2°/sec approximate trim rate

120 sec maximum time required for CONTROL AUG BIT test

6 $\pm$ 1° max rudder trim deflection

FLAP/SLAT System

217 Maximum airspeed for Slats

200 Maximum airspeed for flap extension

50° approximate full flap deflection in degrees

25° approximate 1/2 flap deflection in degrees

15° full slat deflection

SPEED BRAKE System

380 knots speed at which speed brakes blow back begins

340 knots speed above which speed brakes may not fully extend

60° full speed brake deflection

LANDING GEAR System

200 knots maximum gear extension/retraction speed

15 sec approximate landing gear extension time

10 sec approximate landing gear retraction time

2.2-2.5" main landing gear oleo limits (dry)

3.25" approximate nose gear oleo exposed (NATOPS)

7/8-1 7/8" approx main landing gear oleo exposed (NATOPS) (wet)

NOT ON
TEST

0.4 sec back-front seat delay
 100-245 lbs New seat
 136-212 lbs old seat
 600 KIAS and 60K Max seat limits (not body)
 3-4" MDC handle travel

ENGINE SYSTEMS

Rolls Royce F405-RR-401 Turbofan

5,527lbs Thrust

Low Pressure Compressor (N1) - 2 Stage

N1 0-120%

High Pressure Compressor (N2) - 5 stage

N2 0-115%

2 sets of Thermocouples on N1 Turbine

100% N2 - 15,512 RPMs

Ignition engaged until 45% RPM or 45sec

GTS light w/in 20sec on ground

GTS light w/in 28sec in flight

Ready light w/in 15sec

NMM Idle RPM 52+/-2% (before bleed valve closes during 70% climb)

Bleed Valve open w/in 60sec of GTS start

Bleed Valve closes @ 61+/-4% RPM

Increasing N2 RPM 3% and Decreasing EGT 50deg C

Steam ingestion. Bleed Valve opens w/WOW, N2<70%, and Launch bar extended

N2 Accessory Gear Box:

- 1 2 Hydraulic pumps
- 2 28v DC Generator
- 3 LP Fuel Pump
- 4 HP Fuel Pump
- 5 1 Oil Pressure Pump
- 6 3 Oil Scavenge Pumps
- 7 N2 Tach Generator

N2<95% below 20K

N2>90% above 20K

> IN MANUAL FUEL

Ignition deenergizes @ 45% RPM

Generator reset 5sec

72% RPM min approach idle

Idle stop retracts 2sec w/ WOW and Anti-skid off (carrier purposes)

	EGT	RPM	FF
Idle	350-450 deg C	55% +/- 2% +1% per 1.5K	300-400 pph
T/O	x	98-100%	4800 pph
MRT $\dot{C}_{10}$	640 deg C for 30 ^{min} /hr	104%	6410 pph
Acceleration	645 deg C for 20sec	104%	6410 pph
Continuous	550 deg C	100%	x

OIL SYSTEM

MIL-L-23699C or MIL-L-7808G oil

25.2 pints total oil

- C AUG provides yaw damping, ARI, SBI, and rudder trim
- Yaw damping and turn coordination up to 217 KTS
- Yaw damping commands sent through ARI (Aileron, Rudder, Interconnect)
- Engine start switch to ON in both cockpits energizes start control unit
- GTS start button energizes engine igniters for duration of press w/ weight on wheels and duration of press +30 sec w/ weight off wheels (although this doesn't make sense to me because you don't hold the GTS button down while you introduce fuel on a normal ground start, but see p. 26-4)
- From the Engine Start Procedures (Ch 26) know the functions of each button/switch (beginning page 26-3)
- Tail hook is *hydraulically* raised with HYD 1 pressure
- Indications of split slat condition will be uncommanded roll and illuminated SLAT caution light
- Flap position light may not show split flap condition because light is connected to left flap only
- SKID *caution* light illuminates when: failure condition has been detected by anti-skid control box
- SKID *advisory* light illuminates when anti skid selected and landing gear are down and locked
- Paddle switch deselects both NWS and CONTR AUG on the ground. When airborne it deselects CONTR AUG at all times and NWS as long as the gear is down ?
- NWS caution light indicates malfunction within and subsequent shutdown of NWS system or NWS disconnected with paddle switch
- Five times to secure anti-skid switch: after HYD 1 failure (loss of general services hydraulics), after engine shutdown before setting parking brake, carrier landings, anti skid caution light illuminated, blown tire

START

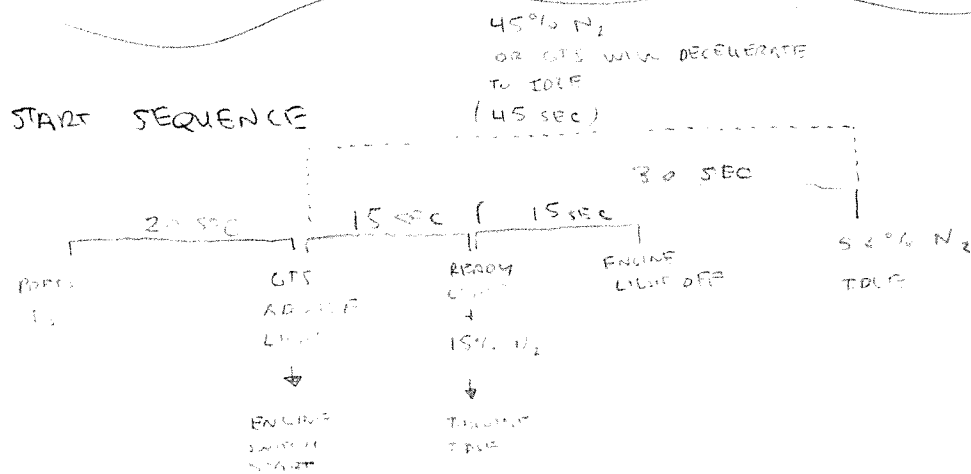
GTS → 20 SEC LIGHT 100%

ENG START SWITCH ON → 15 SEC FOR READY LIGHT

THROTTLE IDLE → 15 SEC FOR LIGHT OFF

BLEED VALVE OPEN - LOOSE

DEFINE DIFFERENCE BETWEEN ABNORMAL STARTS



T-45 Engineering/Systems Gouge

Engine Systems (NATOPS 2.1)

5527 lbs.	Thrust rating of engine
112.4 +/- 1%	N1 rpm above which the EGT/RPM warning light illuminates
104%	Maximum N2 rpm at MRT
104%	Maximum N2 rpm for transient accelerations
100%	Maximum continuous N2 rpm
95%	N2 rpm not to be exceeded in manual fuel control in order to prevent N1 overspeed below 20,000 ft.
90%	N2 rpm not to be exceeded in manual fuel control in order to prevent N1 overspeed above 20,000 ft.
72%	approach idle, minimum rpm gear down and no weight on wheels
61 +/- 4%	% rpm at which bleed valve closes during initial engine acceleration
55 +/- 2%	idle rpm range on standard day (AFTER BLEED VALVE CLOSES)
52%	idle rpm which should be obtained within 30 sec after selecting IDLE
45%	rpm at which ignition is turned off
45%	rpm at which GTS automatically shuts down
45%	approximate rpm at which bleed valve opens during initial engine shutdown
45%	rpm at which ROTATION and GTS advisory lights are extinguished
20%	rpm above which ATS start valve is inhibited from opening TO REDUCE GEAR STRESS ON ATS CLUTCH
15%	minimum N2 rpm required for engine start
3%	approximate increase in idle rpm after compressor bleed valve closes
2%	maximum shift in idle rpm between poststart and engine shutdown checks
1%	increase allowed idle rpm for each 1,500' altitude
650 +/- 8°C	EGT above which the EGT/RPM warning light illuminates
645°C	maximum EGT for transient accelerations → TIME RESTRICTED TO < 20 SEC
610°C	maximum EGT at MRT (30 MIN / HR)
550°C	maximum continuous EGT limit → TIME UNRESTRICTED
550°C	maximum EGT during ground start (NO MORE THAN 550 FOR 10 SEC)
600°C	maximum EGT during air start (NO MORE THAN 600 FOR 10 SEC)
450°C	maximum EGT at idle
300°C	firewire temperature required for FIRE warning light to illuminate
150°C	firewire temperature required for TAILPIPE HOT warning light to illuminate
50°C	approximate decrease in EGT after compressor bleed valve closes
50°C	maximum overshoot above 600°C during airstart (10 sec max)
20°C	maximum overshoot above 550°C during ground start (10 sec max)
30min. / 1 flight hr	maximum time limit at MRT
30min.	minimum interval after three GTS start attempts
3min.	minimum interval between each GTS start attempt
60sec.	Time bleed valve remains open after GTS assisted airstart using manual fuel
45sec.	Maximum time ignition system can remain on → THEN GTS WILL RETURN TO IDLE
45sec.	Time that firewire must remain below 300°C for FIRE warning light to go out
30sec.	Time allowed for GTS to reach GTS idle before automatic shutdown BECAUSE IT GETS TOO HOT
30sec.	Time ignition system remains on after GTS button release during airstart
28sec.	Maximum allowable time for GTS advisory light to illuminate while in flight FOR AIRSTARTS
20sec.	Maximum time for EGT to remain above 600°C during transient accelerations (50°C)
20sec.	Maximum allowable time for GTS advisory light to illuminate at sea level
15sec.	Maximum allowable time for ROTATION advisory light to illuminate READY LIGHT
15sec.	Maximum allowable time engine light-off after moving throttle to idle
10sec.	Maximum time for EGT to remain normal limit during ground start/airstart (FOR 20° OVERSHOOT)
35psi	pressure below which the LP PRESS caution light illuminates
10psi	oil pressure <u>differential</u> below which the OIL PRESS <u>caution</u> light illuminates

Have enough time for Airstart need 1500' AGL and 180 KCAS.

75gph	after 10 sec. Delay.
25.2	additional fuel allowed for acceleration by shot of fuel solenoid below 7200' msl
9.6	capacity of engine oil tank and systems in US pints
0.8	unusable or trapped oil in engine oil system in US Pints
12-joule	maximum consumption rate of engine oil in pints per hour
_____ knots	power rating of two dc ignition plugs
★ 24V	airspeed range for in-flight GTS start attempts
_____ ' MSL	minimum voltage for engine start
9500 +/-300' MSL	maximum altitude for in-flight GTS start attempts
7700 +/- 500 MSL	shot of fuel solenoid closes as aircraft climbs through this altitude
F405-RR-401	shot of fuel solenoid opens as aircraft descends through this altitude
15,512	Model number of Rolls Royce engine
100	N2 rpm at 100% <i>READY</i>
18	the ROTATION advisory light illuminates when N1 rpm reaches tins in the correct direction. Ignition begins too.
5	Number of fuel nozzles in engine
2	number of high pressure (N2) compressor stages
	number of low pressure (N1) compressor stages

Bypass air from the N1 compressor 1. Cools the engine, 2. Provides increased thrust, 3. Provides noise Reduction

T/F Under manual fuel control the compressor bleed valve remains closed at all engine speeds except during airstarts. (TRUE)

5th stage compressor bleed air supplies 1. Environmental system, 2. OBOGS, 3. G-suit, 4. Fuel tank pressurization

Steam ingestion bleed valve provides increased bleed air flow when 1. Weight on wheels, 2. 70% N2, and 3. LAUNCH BAR not retracted

the ACCEL caution light illuminates when 1. Steam ingestion bleed valve not in commanded position (2sec delay), 2. shot of fuel solenoid failure

the engine accessory gearbox drives 1. Hydraulic pumps, 2. DC generator, 3. LP fuel pump, 4. HP fuel pump, 5. Engine oil pumps, and 6. N2 tach-generator

_____ in the FCU dampens throttle valve response during rapid throttle movements.
(Acceleration Switch)

_____ maintains constant pressure drop across throttle valve compensating for airspeed and altitude changes. (Servo Pressure control).

The _____ in the FCU trims basic fuel flow in response to ECA (Solenoid (Fuel Trim))

The _____ automatically prevents overspeed/overtemp conditions by monitoring N1 and two thermocouples behind the N1 turbine. (Engine Control Amplifier)

The _____ meters a fuel supply to the fuel spray nozzles during the start cycle.
(Sub Idle Fuel Control Unit)

T/F The ECA contains two identical lanes, and after engine start, the lane with the lowest fuel flow setting takes priority. (TRUE)

Engine operation in manual fuel control is similar to normal except 2. Airstarts require throttle modulation, 2. Idle rpm up to 5% lower, 3. Slower engine acceleration without dashpot, 4. Bleed valve closed

at all engine speeds except 60 sec on airstart, and 5. N2 must remain below 95/90% below/above 20,000' MSL

With weight on wheels, ignition is provided when 1. GTS button pressed and released, 2. ENGINE switch set to START, and 3. ~~ROTATION~~ ^{READY} is on.

When airborne, ignition is provided when 1. GTS button is pressed and for 30 sec after release, 2. ENGINE Switch set to START, and 3. ~~ROTATION~~ ^{READY} is ON.

The ignition switch has two positions 1. ISOLATE- de-energizes engine and GTS ignition system,
2. NORMAL

Fuel System (NATOPS 2. 2)

MAX FF 6410 PPH

443 gal	total fuel capacity in gallons
427 gal	total usable fuel capacity in gallons
223 gal	Capacity of wing tank in gallons
220 gal	capacity of fuselage tank in gallons
11 gal	amount of unusable or trapped fuel in gallons
3012 lbs.	Total fuel capacity in pounds (JP-5)
2904 lbs.	Total usable fuel capacity in pounds (JP-5)
400 lbs.	Capacity of collector tank in pounds.
350 lbs.	Approximate fuel quantity remaining with FUEL LOW caution light illuminated
73 lbs.	Approximate amount of trapped fuel in pounds
300-400 gph	idle fuel flow in gallons per hour
180 gpm	maximum refueling rate in gallons per minute
24 gpm	defueling rate at 11 psi suction
30 sec.	Capacity of Negative G compartment
30 sec.	Time period 1 seconds fuel boost pumps continue to operate after GEN failure to conserve battery power
10 sec.	Delay in illuminating FUEL LOW caution light to avoid nuisance flickering
50 psi	maximum refueling pressure
9.5 psi	pressure at which dual datum relief valve opens to relieve an over pressure condition
6 psi	pressure of engine bleed air supplied for fuel tank pressurization (Above ambient press.)
4.0 psi	pressure at which dual datum relief valve opens when REFUEL switch set or arresting hook is down
same 3 psi 2 psi 2-3 psi	low air pressure sense switch causes FUEL PRESS caution light illuminates (A/C 163647 and up)
	fuel tank air pressure below which the FUEL PRESS caution light illuminates

Boost pumps automatically turn on 1. During GTS operation, 2. When N2 > 42%, GEN is on, and ENGINE switch is on.

Two fuel boost pumps are installed in the _____. (Negative G compartment)

The FUEL PRESS caution light illuminates when 1. There is insufficient differential pressure across a boost pump (such as an inoperative pump), or 2. The fuel tank air pressure regulator drops below 3 psi.

T/F Opening the fuel shutoff valve disables GST ignition in addition to securing fuel (TRUE)

The fuel flow transmitter is powered by _____ bus and measures fuel flow prior to the _____ pump. (28V Essential Services, LP Fuel Pump)

T/F The FUEL advisory light and the INLET FUEL caution light are inoperative except during the

Light test. (TRUE)

T/F the FUEL LOW caution light is triggered by a float switch in the negative G compartment, and is not linked to the fuel quantity indicator. (TRUE, NATOPS 2.2.2.5)

Fuel tanks are vented to atmosphere through a _____ valve and a _____ valve/orifice. (pressure relief, bleed).

T/F Pressure refueling requires electrical power on the aircraft. (TRUE)

Hydraulic Systems (NATOPS 2.5)

3600 psi	pressure relief valve opens in power supply package (PSP)
3000 psi	normal hydraulic pressure
2500-3000 psi	normal RAT operating range
2200 psi	pressure reserved by wheel brake/emergency flap accumulator if HYD 1 loses pressure
2000 psi	pressure at which HYD 2 PRESS caution light extinguishes
1800 psi	pressure required to close hyd 2 bypass valve and retract RAT if deployed
1660 +/- 110psi	pressure at which HYD 2 PRESS caution light illuminates
1500 +/- 100psi	pressure at which RAT deploys
1600 psi	pressure at which priority valve in PSP reopens to provide power for general services
1500 psi	pressure at which priority valve in PSP closes to isolate flight controls from general services
1300 psi or >	required nitrogen preload pressure for wheel brake/emergency flap accumulator
2600 225 psi	pressure at which HYD 1 PRESS caution light extinguishes
700 psi	pressure at RAT retracts as HYD 2 pressure is lost (aerodynamic load permitting)
1660 +/- 110 600 +/- 50 psi	pressure at which HYD 1 PRESS caution light illuminates
45%	N2 rpm required to use HYD 2 Reset button
42%	N2 rpm below which HYD 2 bypass valve is energized open to reduce engine loads on start
10x	number of full brake applications provided by brake/emergency flap accumulator

three requirements for HYD FAIL warning light 1. HYD 1 pressure below ~~600 +/- 50psi~~ ^{1660 +/- 110}
2. HYD 2 press below 1660 +/- 110psi, and 3. Emergency system pressure less than 600 +/- 50psi

Eight hydraulic devices in general services category 1. Flaps, 2. Slats, 3. Speed brakes, 4. Landing gear, 5. nose wheel steering, 6. Arresting hook, 7. Launch bar, 8. Wheel brakes

_____ ensure adequate base pressure to resist pump cavitation under all flight conditions (nitrogen pressurized resevoirs)

the power supply package consists of _____, _____, _____.
(check valves, pressure relief valve, priority valve)

Flight Controls and Trim System (NATOPS 2.6)

20 +/-0.5°	maximum rudder surface deflection
+6.6° to -15°	maximum travel of stabilator leading edge due to stick movement
+3° to -8°	maximum travel of stabilator leading edge due to trim movement alone
+/- 15.5°	maximum aileron deflection with gear down
+/- 12.5°	maximum aileron deflection with gear up
+/- 9°	maximum aileron trim deflection with gear down
+/- 6°	maximum aileron trim deflection with gear up
2°/sec	approximate trim rate
120 sec	maximum time required for CONTROL AUG BIT test

T/F the artificial feel for the stabilator is provided by the spring cartridge (FALSE, a spring cartridge and inertial weight)

T/F because the standby stabilator trim is powered by the 28 VDC gen bus, in the event of a generator failure, activating the standby trim system will have no effect on the main stabilator trim (FALSE, raising the guard cover over the standby trim switch disengages the main trim motor regardless of the 28 VDC generator bus status)

T/F the stabilator position indicator shows stabilator trim position only with zero force on the stick (TRUE)

T/F rudder trim is available only with control augmentation system activated (TRUE)

The control augmentation system performs four functions 1. Yaw dampening, 2. Turn coordination, 3. SBI interconnect, and 4. Rudder trim.

T/F yaw dampening and turn coordination are not available above 217 knots (TRUE)

T/F ^{ARI} turn coordination is available in any flap configuration (FALSE, only with 1/2 or full)

T/F the SBI operates without any movement of the control stick (TRUE)

The CONTR AUG BIT requires 1. Weight on wheels, 2. Less than 80 knots airspeed, 3. FLAPS/SLATS up

Conditions for CONTR AUG caution light illuminates when 1. CONTR AUG system degraded, 2. BIT test in progress, 3. Paddle switch has been depressed

FLAP/SLAT System (NATOPS 2.7)

217	Maximum airspeed for Slats	MANEUVERING ENVELOPE
200	Maximum airspeed for flap extension	FLAPS/SLATS - +2.0 G
50°	approximate full flap deflection in degrees	- +1.0 G - +1.5 G (ASYMMETRIC)
25°	approximate 1/2 flap deflection in degrees	

T/F assuming sufficient pressure is available, use of emergency flap extension will drive flaps to full down regardless of flap lever (TRUE)

T/F if the brake pressure valve indicates 2200psi or less, activating the emergency flap lever will have no effect (TRUE)

T/F if HYD 1 fails while slats are fully extended, air flow pressure will drive them to the retract position as hydraulic pressure bleeds down (FALSE)

The SLATS caution light is illuminated if 1. Slats not in selected position, 2. Split slats,
3. Slats selected above 217 knots 4. ~~Slats extended then accelerate past 217~~

SPEED BRAKE System (NATOPS 2.8)

380 knots speed at which speed brakes blow back begins
340 knots speed above which speed brakes may not fully extend
60° full speed brake deflection

LANDING GEAR System (NATOPS 2.9)

200 knots maximum gear extension/retraction speed
15 sec approximate landing gear extension time
10 sec approximate landing gear retraction time
2.2-2.5" main landing gear oleo limits
3.25" approximate nose gear oleo exposed (NATOPS)
7/8-1 7/8" approximate main landing gear oleo exposed (NATOPS)

The main landing gear are _____ type units. (Trailing arm suspension)

The _____ caution light is illuminated whenever the landing gear doors are not up and locked (DOOR)

T/F the light in the gear handle will illuminate in the DOWN position if the gear is down and locked, and the gear doors are not up and locked.
(FALSE, gear door status doesn't matter when the handle is down)

T/F The light in the gear handle will illuminate in the UP position if the gear is down and locked, and the gear doors are not up and locked.
(TRUE, gear door status does matter when the handle is in the up Position)

Conditions for WHEELS warning light illumination 1. LDG gear handle is not set to DOWN position,
2. N2 rpm below 95% and either, -
3a. altitude is less than 7200' msl with airspeed below 170 knots, OR
3b. SLATS/FLAPS lever not in up position.

T/F when lowering the gear with the EMER GEAR handle, all of the gear doors remain open
(FALSE, 28 VDC services bus powers the emergency nose landing gear door actuator bringing the nose gear doors to a near closed position.)

NOSE WHEEL STEERING System (NATOPS 2.10)

65° maximum nose wheel deflection with high gain
20° maximum nose wheel deflection with LAUNCH BAR in EXTEND and
NWS button depressed
12° maximum nose wheel deflection with low gain
10 knots maximum taxi speed with high gain NWS

Nose wheel steering is powered off the _____ bus and the HYD 1 system (28VDC Essential Services)

T/F On landing, the NWS system is engaged with weight on wheels from just one main gear and nose wheel, and will remain engaged with weight on only one main gear (TRUE)

T/F In order to use NWS with LAUNCH BAR in Extend, the NWS button must be depressed (TRUE)

The NOSE WHEEL STR caution light is illuminated if 1. Nose wheel moves away from commanded

pedal position, 2. If there is an internal system failure (including HYD 1 failure), 3. When the system has been paddled off.

T/F No landing gear will retract until the nose gear is centered (TRUE)

The paddle switch 1. Disengages NWS, 2. Deactivates control augmentation causing the NOSE WHL STR and CONTR AUG caution lights to illuminate when the aircraft is on the ground

T/F ~~the NOSE WHL STR~~ ^{NWS HI} advisory light is illuminated anytime the NWS system is energized (FALSE, only shows on high gain)

WHEEL BRAKES/ANTI-SKID System (NATOPS 2.11)

2200psi	pressure reserved by wheel brake/emergency flap accumulator if HYD 1 loses pressure
180 +/-20psi	minimum pressure to illuminate BRAKE PRESSURE indicators lights
150psi	pressure at which BRAKE PRESSURE lights are extinguished
30 knots	minimum required wheel speed for anti-skid to operate while accelerating
10-13 knots	minimum wheel speed for anti-skid to operate while decelerating
10	number of full brake applications provided by brake/emergency flap accumulator
324°F	temperature at which fusible plugs in main gear wheels are designed to release tire press.

When landing gear is retracted, landing gear door open pressure is applied to the _____ cylinder. Which in turn applies force to the parking brake lever to actuate the brake control valve (DE-SPIN)

Conditions for anti-skid to operate 1. Landing gear down, 2. Weight on wheels, and 3. Speed above 30 knots while accelerating or 10-13 knots deceleration

T/F the anti-skid system releases brake pressure if brake pedals are depressed while airborne to prevent touchdown with locked brakes. (TRUE)

the ANTI-SKID caution light is illuminated if 1. System malfunction is detected, 2. Prolonged full pressure dump lasting greater than 2 seconds, 3. No wheel spin detected 3 seconds after weight on wheels (reverts to normal braking)

T/F engaging the parking brake with anti-skid energized will eventually deplete brake accumulator pressure (TRUE)

T/F selecting anti-skid in either cockpit will energize the anti-skid system (FALSE, both cockpits must have it selected)

The PARKING BRAKE caution light is illuminated if the throttle is advanced beyond intermediate position (60-70%) and the parking brake is engaged.

Conditions for ANTI-SKID advisory light illumination 1. Anti-skid switches in both cockpits must be ON, and 2. Landing gear DOWN

LAUNCH BAR System (NATOPS 2.12)

10 sec. Time allowed for launch bar to retract following launch before warning tone sounds

The green L BAR indicator light is illuminated when launch bar is extended with the switch set to EXTEND

The red L BAR indicator light is illuminated when 1. Launch bar is not retracted, 2. No weight on wheels, 3. gear down and not locked

T/F with a launch bar retraction failure, raising the gear is not possible. (FALSE)

T/F the weight on wheels switch inhibits launch bar switch from remaining in extend position when airborne (TRUE)

ARRESTING HOOK System (NATOPS 2.13)

950psi +/-50 pressure of arresting hook nitrogen preload
300 knots airspeed above which HOOK warning light may illuminate due to airloads
6 sec approximate hook transit time up
1.5 sec approximate hook transit time down
the HOOK warning light is illuminated when HOOK handle does not correspond to hook position

T/F moving the HOOK handle to the down position hydraulically releases the up latch assembly and switches a hydraulic selector valve to remove HYD 1 pressure (FALSE, it mechanically moves both)

BOARDING System (NATOPS 2.14)

T/F both pull-out footsteps can be retracted from inside the cockpit (FALSE, front only)

CANOPY System (NATOPS 2.15)

95% rpm above which caution tone sounds with CANOPY caution light illuminated
45 knots maximum side winds allowed for canopy to remain open
4" free travel range of MDC firing handle

EJECTION SEAT System (NATOPS 2.16)

18,000' MSL altitude at which drogue bridles are released during Mode 5 ejection sequence
0.5 sec backup delay initiator between seats
0.4 sec interseat sequencing system delay
30-60 lbs. Force required to initiate ejection
60-70% rpm above which SEAT UNARMED caution light will illuminate

ENVIRONMENTAL CONTROL System (NATOPS 2.18)

150°F (67°C) temperature at which AVIONIC HOT caution light illuminates with weight on wheels
5 sec time in seconds between each rotary valve movement in oxygen concentrator
60/40 approximate airflow split between ventilation and defog with airflow knob in normal

Normal 60/40 Below 40/60

T/F in the event of complete electrical power loss, air conditioning and pressurization will remain on (TRUE, the PRSOV IPRSOV valves deenergize to the open position)

The ram air inlet and outlet valves are open when 1. Air flow knob is in the off position 2. W-O-W

T/F the AVIONIC HOT caution light is only operative on the ground (TRUE)

Conditions that will cause CABIN ALT warning light to illuminate 1. Air conditioning failure causes overpressure to the CAU compressor 2. The CAU compressor outlet temperature rises above 500°F, 3. The CAU turbine inlet temperature rises above 250°F cabin pressure failure only when cabin altitude is above 24,500' +/- 500'

ON-BOARD OXYGEN GENERATING System (NATOPS 2.19)

5 sec time in seconds between each rotary valve movement in oxygen concentrator
250°F temperature at which overheat temperature sensor illuminates OXYGEN Warning light
40000' MSL altitude above which full 4.0 psi cabin pressure is obtained
35000' MSL cabin altitude above which oxygen is supplied at full pressure
35000' cabin altitude above which oxygen is supplied at increased pressure
24500 +/-500' cabin alt above which CABIN ALT warning light will illuminate
9500' MSL altitude above which a low oxygen concentration will cause OBOGS to shutdown

OVERHEAT - OVERPRESSURE OF COLD AIR UNIT

$$\text{CABIN ALT FORMULA} = \left(\text{PAIR} \div 2 \right) + 3000$$

5000' MSL	altitude at which pressurization begins
2 G	number of G's at which anti-g suit inflation begins
6 G	number of G's at which anti-g suit reaches maximum inflation
1800-2500 psi	acceptable pressure range for emergency oxygen bottle
6.5 psi	maximum pressure of anti-g suit inflation
4.8 psi	cabin pressure required to open safety relief valve
4 psi	maximum cabin pressure allowed by discharge valves
-0.5 psi	pressure below which safety and inward relief valve opens
4-20 min	approximate duration that emergency oxygen supply provides in minutes

$$\text{EX// } \left(\frac{14000}{2} \right) + 3000$$

$$7000 + 3000$$

$$\text{CABIN ALT} = 10,000'$$

* NEED O₂

three situations that will cause OBOGS automatic shutoff 1. A system malfunction 2. Low oxygen concentration and altitude over 9500' and, 3. Bleed air temp above 250°F or 4. Whenever the OBOGS/ANTI-G switch is off

T/F Once started, the emergency oxygen supply cannot be shutoff and restarted (FALSE)

FLIGHT INSTRUMENTS (NATOPS 2.20)

5°	degrees of deviation indicated by increments on the CDI for VOR and TACAN
1 1/4°	degrees of deviation indicated by increments on the CDI for ILS (usually)
9 min	2 1/4' standby gyro provides valid attitude data for minimum of __ after power loss
3 min	3" standby gyro provides valid attitude data for minimum of __ after power loss
399.9 nm	maximum range that can be displayed on H.S.I. range indicator
30	number of preset channels available
000	H.S.I. course required for VOR/ILS self test
180	H.S.I. course required for TACAN self test
315 +/- 2.5°	heading No. 1 points to during VOR/ILS self test
180 +/- 2.5°	heading No. 2 points to during TACAN self test
108.10-111.95 MHz (odd)	ILS frequency range
108.0-117.95 MHz	VOR frequency Range
_____	-173.975 MHz VHF frequency Range

Static inputs are provided to 1. Mach/airspeed indicators, 2. Barometric altimeters, 3. VSI's 4. OBOGS, 5. SADS

T/F The Mach indicator is corrected of temperature and pressure (TRUE)

T/F the forward and rear mach/airspeed indicators contains an airspeed switches set for 170 and 217 knots respectively to signal appropriate cautions and warnings (TRUE)

the SAHRS system provides only heading information to the H.S.I.

the SAHRS system provides only roll information to the yaw damper control (YDC)

The SAHRS system provides attitude information to the 1. ADI, 2. HUD, and 3. ADR

T/F The ILS needles come into view on the ADI whenever a valid ILS frequency is tuned (FALSE, ILS must be selected on NAV/COMM control transfer panel)

T/F The turn indicator on the ADI will be available in the event of a GEN failure (FALSE)

T/F the number 1 green needle corresponds to VOR and number 2 white needle to TACAN (TRUE)

T/F When T/R&G is selected on the COMM radio, both 121.5 and 243.0 MHz are monitored (FALSE, only the GUARD freq in the selected band)

ANGLE-OF-ATTACK System (NATOPS 2.21)

25.5-28 units	AOA Range for Stall
21.5 units	AOA at which rudder shakers activate
>18 units	AOA range for upper green chevron (V)(Slow)
17 1/2 -18 units	AOA range for upper green chevron and amber donut (VO) (slightly slow)
17 units	optimum AOA, on speed
16-16 1/2 units	AOA range for lower red chevron and amber donut (^O)(slightly fast)
< 16 units	AOA range for lower red chevron (^)(fast)
14 units	AOA for best angle of climb and max endurance (holding), ~195-210 knots
12 units	AOA for best rate of climb and max range (Bingo)
T/F	the AOA indexer lights illuminate with weight off wheels and the gear handle down (FALSE, need three gear down AND LOCKED)
T/F	discrepancies are possible between rear cockpit AOA indicator and rear cockpit indexer lights (TRUE)

RADAR ALTIMETER System (NATOPS 2.22)

5,000' AGL	maximum altitude displayed on radar altimeter
100 +/- 10'	altitude indicted during the BIT check
40°	maximum pitch or bank angle for radar altimeter effectiveness
AN/APN-194	Radar altimeter designation
T/F	the low altitude warning system (LAWS) can be set in either cockpit, but the headset warning tone will only be triggered by the front cockpit radar altimeter. (TRUE)



T-45C ENGINEERING GOUGE

1. What two capabilities does the Display Electronics Unit possess?
Mission Computer and Display Computer
2. T/F The Aft MFDs operate through the forward MFDs.
False. They operate independently.
3. In what order does the Auto BIT Operate?
DEU, GINA, ADR, MDL, RADALT
4. What is the function of the MIL-STD-1553B Multiplex Data Bus?
DEU interface to ADR, MDL, and GINA
5. In that format does information travel through the MUX Bus?
Digital
6. How many push-button options are on each MFD?
20
7. Name the major components of the Display System.
DEU, HUD, 2 DEPs, 4 MFDs
8. Name the four types of GINA BITs.
Power up, Initiated, Continuous, Manual
9. What type data do the Hard-Wired interfaces transmit between the DEU and their components?
Analog, video, discrete
10. What is the solution to one MFD experiencing a display problem?
Multiple MFDs with the same problem?
Turn off the MFD
Display Power switch momentarily to reset (cycles power to DEU)
11. Name the major components of the GINA
GPS and INA
12. T/F For the H.S.I. to fully work, you need essential and non-essential power.
True, you need both to run the VOR/ILS
13. With a generator failure, batteries will give _____ minutes of power.
27
14. Overvoltage unit takes generator offline if output exceeds _____.
30.2 VDC
15. Undervoltage unit deenergizes non-essential buses when output drops to _____.
25 VDC
16. What are the indications of an overvoltage condition? Undervoltage?
OVERVOLT: GEN Warning LT, ~~CAUTION LT (30 sec)~~, CAUG
Caution LT, RMFD disabled, FPRES Caution LT (>30 sec), HUD fail.
UNDERVOLT: GEN WARNING, ~~CAUTION LT (30 sec)~~
17. How many essential buses are there? Non-essential?
Three (2 AC, 1 DC) for both.
18. What is the ~~GEN~~ Idle RPM?
86%
GTs
19. T/F Losing one of the ECA lanes will illuminate the ECA Caution LT.
False. Losing one lane will illuminate the ECA advisory light.
20. Describe the verifications of an ECA failure (Full Trim).
RPM decreases, EGT decreases, FF decreases
21. Describe the verifications of an ECA failure (NO Trim).
RPM increases, EGT increases, FF increases
22. At what quantity of fuel does the FUEL Caution LT illuminate?
350lbs
23. T/F The FUEL Caution light and the Fuel Gauge receive info from the same float switch.
False. The FUEL Caution LT has its own float switch.

2 I missed

• Fire light is first indication of FIRE during T/c

• What are secondary indications of fire? All of above

TRUE

HYD FAILURES

3000 PSI NORMAL FOR ALL

HYD 1

3000 PSI - NORM

HYD 2

3000 PSI - NORM

1660 $\pm$ 110 - CAUTION LIGHT

1500 PSI - RAT EXTEND

2500-3000 - RAT POWER

2000 - LIGHT EXTINGUISHED

1800 - REQUIRED TO RESET RAT

24. What are the verifications of a steam ingestion valve failure?
OPEN - RPM lower, EGT higher, decreased thrust and acceleration
CLOSED - Engine surge/stall due to steam ingestion (cat shot)

25. On engine start, when does the GTS shut down?
N2 = 45%

26. When does the HYD 2 Caution LT illuminate? When does it extinguish?
HYD 2 EDP below 1660 +/-110
HYD 2 EDP above 2000

27. When does the RAT deploy?
HYD 2 EDP below 1500

28. When does the HYD 1 Caution LT illuminate?
HYD 1 EDP below 1660 +/-110 (~~600 +/- 50 before AFC 244~~)

29. What is first indication of HYD 1 slow pressure loss?
1500 psi, loss of general services (no light before AFC 244)

30. When can the RAT be reset? How will you know it reset?
1800 psi HYD 2 EDP pressure
HYD 2 Caution LT will extinguish at 2000 psi HYD 2 EDP pressure

31. T/F The Wheel brake/ Emergency Flap Accumulator allows for 10 wheel brake applications and Flap/Slat extension in the event of HYD 1 failure.

False. Allows for 10 wheel brake applications and Flap extension only (no Slats)

32. At what pressure will the Wheel Brake/Emergency Flap accumulator shutoff access to the flaps?
2200psi

33. How and when is the HYD 2 bypass valve closed after engine start?
HYD 2 RESET button once engine is above 45% N2

34. When does the WHEELS Warning LT illuminate?
a) Altitude less than 7200' MSL and airspeed less than 170kts or;
b) FLAPS/SLATS lever not in up position

35. Where is the ANTI-SKID switch located?
Left console, both cockpits

36. T/F If one ANTI-SKID switch is off, the anti-skid system will not operate.

True.
37. Name the general services lost when HYD 1 drops below 1500 psi.
-Normal Gear Extension
-Normal FLAPS/SLATS
-Normal Brakes
-NWS
-LBAR extension
-Speed Brake
-Hook Retract

38. When does SLATS Caution LT illuminate?
Slats not in selected position

Split Slats
Slats extended and airspeed increases above 217 kts
Slats selected above 217 kts (slats won't extend)

39. What is the function of the paddle switch on the stick?
Disengages NWS and Control Augmentation

40. When does the SKID Caution LT illuminate?
Failure condition detected by anti-skid control box.

41. When does the SKID Advisory LT illuminate?
ANTI-SKID selected and gear down and locked

42. How do you disengage the CAUG system? Reengage?
Paddle Switch on stick
CAUG Switch

43. What are the three positions on the CAUG switch and functions?
ALL - Dutch Roll, ARI, SBI, Rudder Trim
SBI - Rudder Trim, SBI

- Failure of bleed valve to close will cause loss of thrust at M2T.
- Anti-SKID switches: ONE OFF then the system is off
- Loss of one inverter will light AC INV caution light and loss of AC non-ess.

Cont Aug Sys

W(ON)W:- SBI ; Rudder trim

W(OFF)W:- SBI ; Rudder trim

- ARI (Aerodynamic Rudder Trim)

- YAW Damping $\downarrow$ 217

Control Aug Sys

1) ALL: ^{① (Yaw Damp)} Dutch roll damp, ^{② (ARI)} Turn Coord, ^③ rudder trim.
- ^④ Speed brake stabilizer, Intereconnect (SBI)
* ^⑤ Initiates Bit (WOMW, 280 bits
Flaps/Slots on up.)

2) SBI: ^① Rudder Trim ; ^② speed brake stab Intereconnect (SBI)

3) RESET: momentary flen to SBI
~ Resets system

RESET - resets system

44. What is the function of the SBI system?
Provides stab trim inputs to reduce pitch changes during SB ext.
45. What is the function of the IPRSOV?
Cooling air inducted through heat exchangers when gear extended
46. What happens when the ADR button is depressed?
ADR retains flight incident for -5 and +5 seconds.
47. What is tested when the light test switch is put to Test position?
Tests all lights minus AOA Indexers and tests Fire Detection Sys.
48. List the Warning lights on the CWS panel.
OIL PRESS, CABIN ALT, EGT/RPM, FIRE, GENERATOR, GTS FIRE, HYD
FAIL, OXYGEN
49. When you switch the light test switch to LAMP, how will the fuel quantity system be tested?
It won't. (Trick question: FUEL Lt is a Caution Lt, not Warning)
50. What is the max EGT for airstart (excluding overshoot)?
600 deg
51. The Engine Switch controls power to the _____.
SCU (Starter Control Unit)
52. T/F With a complete electrical failure, Landing Gear and Flaps may be extended manually.
False. Only the gear may be extended manually.
53. What is the affect on the Tailhook with a HYD 1 failure?
May be lowered, but not raised.
54. T/F A single AC Inverter failure will result in AC INV Caution LT.
True.
55. What Trim functions are on Generator Bus?
Aileron Trim, Rudder Trim, STBY Stab Trim, Trim Indicators
56. How is generator voltage regulated?
Voltage regulator: 28.2 - 29.2 volts
57. List 4 generator protection systems.
Voltage regulator,
Overvoltage protection unit (turns gen off at 30.2 volts)
Undervoltage protection unit (deenergizes Non-essen AC if <25 v)
★ → 300 Amp fuse ←
58. If Overvoltage or Gen Fail, what happens?
Gen Bus - Off
Non-essential AC Buses - Off
Generator Warning Light
Master Caution
FPRESS - On (after 30 sec)
CAUG - On (CWS tone also)
HUD - Out
RT MFD - Out
LT MFD - Reverts to ADI. Out after 2 minutes unless switch ORIDE
59. If Undervoltage, what happens?
Generator Warning Light, A/C non essl drops
Master Caution, volts read 24
60. Will low generator output drain batteries?
No. Power diodes protect current from being drawn from Essen Bus
61. What powers the 28 VDC Essen Service Bus?
24 VDC Batts and Gen
62. How much time is available on Battery power alone?
27 minutes on two good batts
12 minutes on one good batt
63. Ground Service Bus runs off of what?
#1 Battery

- Illumination of GEN light with undervoltage OR overvoltage.
- OXYGEN Light - OBAGS still works just below 95% oxygen

64. What happens if lose one inverter?
ACINV caution light and lose non-essential AC busses
65. What happens if lose two inverters?
ACINV caution light and lose all AC ^{per}
66. How is AC power reset for input faults? Output faults?
Input faults is auto-reset.
Output faults requires manual reset (cycle) *(8 sec (not needed) for reset)*
67. With one Inverter failed, what means of navigation do I have?
TACAN azimuth.
68. With two Inverters failed, what means of navigation do I have?
PAR approaches only.
69. What are the vital services lost when non-essential AC busses lost?
VOR/ILS/MB and RADALT
70. TACAN DME is off of what bus?
28 VDC Essential
71. What is the biggest draw of electrical power?
Pitot Heat
72. What should I expect if I see a Master Caution light, tone, loss of RT MFD and HUD?
Generator Failure
73. What controls engine from start to finish?
Start Control Unit (SCU)
74. If either cockpit is in Manual Fuel, engine is operating in ____.
Manual Fuel
75. If finger lift is raised airborne with throttle at approach idle stop, further reduction will cause what?
Shutting the engine down.
76. If finger lift is raised airborne with throttle above approach idle stop, further reduction will cause what?
Throttle will stop at 72% (Approach Idle)
77. What is the minimum RPM for Engine Start?
15%
78. What is the minimum RPM for Air Start?
13%
79. When does the OIL PRESS light go out on start?
18%
80. What does the SLAT Caution light indicate?
Slat are not in commanded position or are unsymmetric *slat out A 217*
81. What is the airspeed limitation on the Flaps and Landing Gear?
200 kts *slat su used A 217 - no slatops*
82. What constitutes the ARI system? *(Pg 13-23)*
Yaw dampener and trim tab system
83. What drives the Rudder Centering system?
Spring
84. If I get a CAUG caution light, what do I lose?
ARI (217 kts, Flaps half or full), SBI, Rudder Trim, Yaw Dampening (<217kts)
85. CAUG light with a Generator Failure, I lose what?
ARI, SBI, Yaw Dampening, Rudder Trim (ALL FOUR)
86. CAUG Bit can only be conducted when?
<80kts; WoW; Flaps up
87. In SBI ~~with~~ front SBI rudder trim still work?
Yes *will*
88. What navigation do you have with a DEU failure?
None. Only option is the PAR
89. GINA supplies what flight information?
Velocity Vector, Gyro, Heading, VVI, *position*

PR BRAKE CAUTION LIGHT - ON IF SET AND N₂ ABV 70% RPM.

• GEN failure - only trim left is primary stabilator

90. Pitot Static provides what flight information?
Airspeed, Barometric Altitude
91. What page is initiated on power up?
Menu
92. How do I know when alignment is completed?
Velocity Vector appears in HUD
93. What are the two types of Bit Displays?
WoW and WoffW
94. What does OPGO mean.
Only one channel of MUX Bus is Tx properly
95. What must be done to display GS/LOC needles?
Box ILS on the H.S.I. (must be in CDI not Planometric)
96. What are the CDI deflections? TCN 10-radials VOR 10-bearing lines
~~10-radials~~
4 miles for CL on WYPT/WOS Gear Up
.3 miles for CL on WYPT/WOS Gear Down
2.5 degrees for LOC (Limits of ILS usually 2.5)
97. IFF has _____ antennas. Which antenna signal is selected? in Div mode
Two. Strongest.
98. Which IFF Mode transmits altitude encoding?
Mode 3C
99. What is the normal distribution of ECS airflow?
60% for ventilation and 40% to Defog
100. When and what does IPROV do?
With gear down, improves ECS efficiency
101. Ram Air opens with what?
WoW
102. What does AV HOT mean? (50°F)
ECS cool temp exceeds 67 deg C. Must cool before takeoff
103. What is the airflow distribution in Defog?
40% Ventillation and 60% Defog
104. What does Max Defog do?
Increases Airflow at same distribution ration (60/40)
105. What does CABIN ALT Light mean?
Cabin Altitude exceeded 24,500 +/-500 or Overpress/Overheat of Cold Air Unit (CAU) of 500 degree on compressor outlet or 250 degrees on Turbine Inlet.
106. What does OXYGEN light mean?
OXYGEN switch off, O2 concentration below 95%, exceeds 250 degrees at temp switch.
107. If MDC fails, will seat still breach canopy?
Yes, canopy penetrators will fracture canopy glass.
108. Where do the inputs for ejection modes come from?
Ejection sensor on seat
109. What does CANOPY light mean?
Canopy latch is not in the latched and locked position.
110. When do you get the CANOPY light and MASTER CAUTION light?
With the canopy unsafe and the power set to above 95%
111. Electrical boost pumps are located where? What function do they provide?
Located in the negative G compartment of the collector tank.
Provide fuel pressure for negative G flight.
112. LP and HP pumps are powered by what?
External Drive Gear Box
113. If FIRE light comes on and goes out, what should be done?
Check light switch for continuity
114. In event of HYD failure, will ANTI-SKID or NWS lights illuminate?

CABIN ALT WARN LIGHT - Pg 17-19

AV HOT LIGHT - 76.7°C ONLY ON GROUND

- No. They will only illuminate with electrical system malf.
115. ACCEL light means what?
Steam Ingestion Valve stuck open (10% loss of thrust), or Initial Shot of Fuel Solenoid is not in commanded position
116. What is the function of the Sub-Idle Fuel Control Unit (SIFCU)?
Deliver proper pressure and pattern to nozzles.
117. External Gear Box drives what?
2 Fuel Pumps (LP/HP), 2 HYD EDPs, 4 OIL Pumps (3 scavenge), *rev*
118. What is the min airspeed / altitude to initiate restart after flameout?
1500' / 180 kts
119. What is the prime indicator of engine seizure?
RPM rapidly decays to zero.
120. How do you determine that the engine is locked in stall?
EGT over 450 for 6 seconds after selecting idle
121. What does EGT/RPM light mean?
EGT > 650 +/-8 or RPM 112.4 +/-1.0 (N1 RPM)
122. How do you shutoff fuel to Engine and GTS?
Fuel Shutoff T handle
123. What does FIRE LIGHT mean?
Temp outside engine container exceeds 300 deg C
124. What does GTS FIRE mean?
Temp outside GTS container exceeds 300 deg C
125. What does TP Hot Caution Light mean?
Temp outside tail cone inside fuselage can exceeds 150 deg C
126. What does HYD 1 supply hydraulic power to?
General Services and Flight Controls
127. What does HYD 2 supply hydraulic power to?
Flight Controls
128. What is the indication of an operable RAT?
HYD 2 fluctuating between 2500-3000
129. If HYD 2 and RAT fail, what effect on total HYD system?
None. HYD 1 still operating everything normally.
130. If HYD 2 is leaking, will RAT do the job.
No.
131. HYD 1 light on below _____. Resets at _____.
1660 +/-110. 2000.
132. What pressure is required to reset HYD 2?
1800 psi
133. What is the Nitrogen preload of the Emer Brake/Flap sys?
1300 +/-50
134. If Brake Accum drops below 2200 psi, what happens?
Emergency Flap accuation is removed to isolat Emer Brake
135. Brake Accumulator gauge is reading what?
Nitrogen pressure
136. Flight control accumulators have precharge of what?
1100 +/-50 psi nitrogen
137. When is ANTI-SKID activated and deactivated?
Accelerating above 30 knots and decelerating below 10-13 kts *above*
138. With HYD 1 failure, does hook work?
~~Yes. Pneumatic and snubber to put and lock down. (Can't raise)~~
139. ANTI-SKID off, carrier landing, and HYD 1 fail: *5*
Blow tire, parked, ~~with~~ light on *(Anti)*
140. LBAR is raised and lowered how?
Lowered by HYD 1. Raised by mechanical spring.
141. If the FC has control of the CNI systems _____
Answer: *Front cockpit* ...the RC can select any MFD display within the system *rear cockpit*

system fail

① failed Brake press. release system

Locked In saddle for launch
① G & i Coiled
② LBAR & i locker, SW Retract
③ Throttle $\geq 95\%$
④ NWS not passed
⑤ ph brake off component circuit failure, IV pit.
Indicator circuit failure

142. Paddle switch with gear down does what?
Switches off the NWS and the CAUG
143. To reset NWS and CAUG, you must do what?
NWS: depress the NWS button
CAUG: momentary select to CAUG reset
144. With NWS on, what will activate the NWS?
WoW
145. BRAKE PRESS light comes on when?
LEAR not retracted, PBRAKE not set, over 95%. Etc. (know all)
146. If DEU fails, what navigation aids do I lose? What are my options?
Everything. PAR or Visual
147. DEU Controls the MUX BUS (that's the best answer)
148. DEU exchanges _____ words with _____, _____, and _____ via MuxBus.
Digital. ADR, MDL, GINA
149. How do the hard wired interfaces communicate?
Analog, video, discrete
150. What page is initiated on power-up?
Menu page
151. Display system derives flight data from GINA and Pitot Static inputs. Which are from each?
GINA: pitch, roll, heading, vertical velocity, position ⑤
Pitot: altitude, indicated airspeed
152. What MFD inputs are duplicated on the DEP?
LAW, CRSE, HDG, ENG ①
153. T/F GINA is operational upon loss of generator.
True
154. Auto Initiation of IBIT is accomplished in what order?
DEU, GINA, ADR, MDL, RALT
155. What are the two BIT Display pages?
WoW and WoffW
156. TACAN uses _____ bearing while VOR uses _____.
Magnetic. Relative
157. IFF Mode Test illuminates GO/NOGO light or ANTENNA lights?
GO/NOGO Lights
158. IFF DEGRD on the BIT page indicates what?
1. IFF - Off, or
2. IFF - STBY, or
3. IFF - BIT failure
159. What IFF mode transmits altitude?
Mode 3C
160. The VCR records what?
HUD, MFD, Tones, CWS, ICS, Radio ⑥
161. T/F Undervoltage protection ALWAYS trips Non-essen AC Bus Off
True
162. With GEN FAIL, will ALL flight controls and sub-systems work?
Yes. Only lose ail/rud/stby stab trim, trim indicators, HUD, RT MFD, and #2 UHF/VHF
163. T/F Approach Idle function engages with total electrical failure.
True. Normally held open magnetically.
164. What AOA will Stall Warning Indications occur?
21.5
165. How do I know when alignment is complete?
Velocity Vector appears in HUD
166. T/F Advisory Windows appear on MFDs and HUD?
False. Only appear on MFDs
167. Engine Bleed Valve closes at _____ % RPM.
61 +/-4

168. Describe Hot, Hung, Wet starts.

Hot: FF and RPM to normal Idle indications, but EGT rapidly to 550 or to 570 for more than 10 seconds

Hung: RPM stagnates IVO 45% but EGT continues to rise

Wet: FF rises normally, but no EGT/RPM within 15 seconds of idle

★ 169. GEN Warning light extinguishes on start at ___, comes on at ___ on shutdown.

45% 42%

★ 170. OIL PRESS light goes off at ___ on engine start.

18% (18-20)

★ 171. Selecting ANTI-SKID off and CARRIER mode does what?

Provides 2 second delay for Approach Idle upon WoW

172. GTS light must appear within ___ seconds of initiating GPS start.

20

173. Must have READY light within ___ seconds of selecting start.

15

174. What does READY light mean?

GTS at full speed, N1 at 100% ^{RPM} in right direction, IGN. ON

175. Start Limits:

FF: 300-400

EGT: 550 (570 for 10 sec)

RPM: 52 +/-2 (within 30 seconds of idle)

176. Idle limits:

FF: 300-400

EGT: 450

RPM: 55 +/-2 (1% for every 1500')

177. Engine Limits:

FF: 6410 max

EGT: 610 max @ MRT

550 continuous

645 transient (20 seconds)

600 airborne start (650 overshoot 10 seconds)

% RPM: 104 max (less than 30 minutes per flight hour)

104 transient (20 seconds max)

100 max continuous

55 +/-2 idle

★ 178. GINA uses GPS/INS with what software?

Kalman filter (software -NOT System)

? 179. Initial Shot of Fuel Solenoid is actuated when? Off when?

Descending through 7700 +/-500' MSL

Ascending through 9500 +/-300' MSL

180. When is RAM air activated?

With WoW.

HYDs

HYD 1 Numbers:

- 3000 psi at 9.6 gpm - EDP Out
- 1100 +/- 50 psi - FC Accum Nitrogen Press
- 1300 +/- 50 psi - Brake/Flap Accum Nitrogen Press *emerg*
- 1500 psi - Priority Valve - No General Services
- ~~1600~~ 1600 psi - Priority valve redirects to General Services
- 2200 psi - No Flap/ Just Brakes
- 1600 +/- 50 - HYD 1 Light
- 725 +/- 50 - HYD 1 Light Out

HYD 2 Numbers:

- 3000 psi at 6 gpm - EDP Out
- 1100 +/- 50 psi - FC Accum Nitrogen Press
- 1500 psi - RAT Deploys, Deployed by EPA
- 1800 psi - HYD 2 Reset Allowed by EPA
- 1660 +/- 110 psi - HYD 2 Light
- 2000 psi - HYD 2 Light Out
- 2500-3000 psi - Normal RAT Output
- 700 psi - RAT Retract (? Airloads)
- 7300 rpm - RAT Governor Maintains
- 42% ↓ - Bypass Effective
- 45% ↑ - Bypass Accepts Reset

EPA - EMERGENCY
PACKAGE
ASSEMBLY

RESET HYD 2

Brake/ Flap Accum allows for Flap Emerg Ext and 10 Brake applications

HYD Fail Warning - HYD 1 < 600 +/- 50, HYD 2 < 1660 +/- 110, Emerg Sys < 600 +/- 50

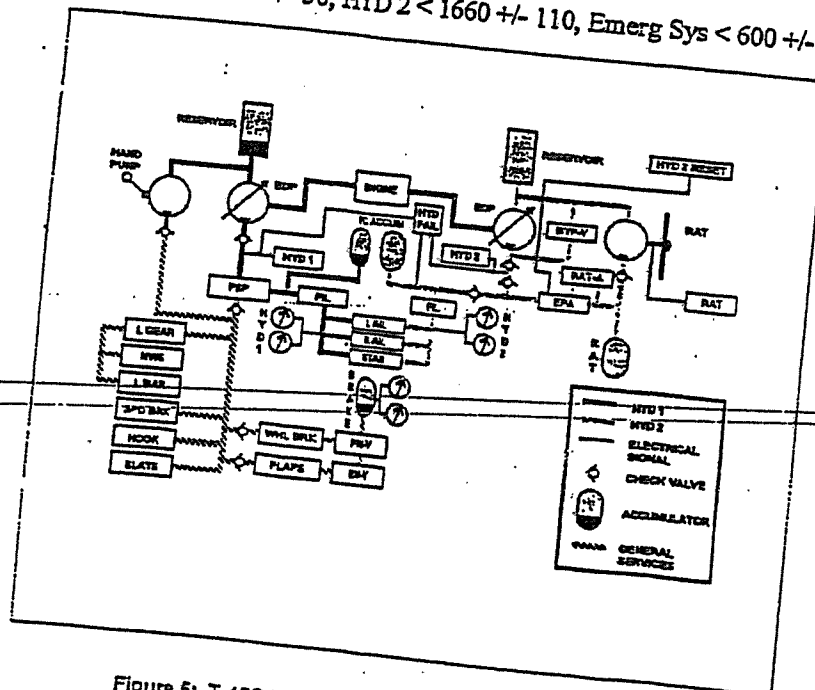


Figure 5: T-45C HYDRAULIC SYSTEM BLOCK DIAGRAM

GROUND {
TAIL PIPE HOT - EXEC. EMERG SHUTDOWN
FIRE - EXEC. EMERG SHUTDOWN
CTS FIRE - EXEC. EMERG SHUTDOWN (I/F ENG SWITCH OFF, SECONDARY INDICATION)

AIRSTARTS

IMMEDIATE - ANY ALT

WINDMILL - < 25000' MSL, 13% N₂ MIN

ASSISTED - < 15,000' MSL, < 20% RPM - CTS LT ON

EAGLE EP GOUGE

EP 06X

1. If your HYD 1 and 2 fail but the RAT is working fine, but the aircraft is "uncontrollable", then EJECT.
2. Engine Stall = Throttle Idle, Controls Neutralize, EGT/RPM Monitor.
3. Ready Light = N1 @ 100 RPM in correct direction with ignitors energized.
4. If you are flying form and prove that you can no longer fly form together (YOU HIT EACH OTHER), then have someone else check you out (besides the guy that you just hit).
5. Power setting for OIL PRESS light = min for safe flight. / MINIMIZE THROTTLE MGMTS
6. Engine Failure off Cat = EJECT (if unable then ditch straight ahead).
7. TP HOT caution light on deck is same as FIRE light on deck (Same procedures to deal with both).
8. Configuration for ditching = gear up flaps down full.
9. For blown tire... turn anti-skid off.
10. If you lose nosewheel steering, use differential braking to control aircraft.
11. If you get an overspeed during start, use Clear Engine procedure (Throttle Off is the first step... not Throttle Idle).
12. There are two GTS Fire questions with the same answer... Engine Switch Off is the first step.
13. Which of these indicate you might have an engine fire on startup? Fire light, Increased FF, Rapidly increasing EGT? ALL OF THE ABOVE.
14. With a HYD 1 Failure, which of these can you do for landing? Emergency extend the gear, Emergency extend the flaps, Take an arrested landing? ALL OF THE ABOVE.
15. With a HYD 2 failure, you monitor the HYD 1 PRESSURE GAUGE not the accumulator pressure.
16. Compressor Stall first step is Throttle Idle – min for safe flight.

17. Compressor Stall (or engine stall) is characterized by "pops and bangs" and an increase in EGT and decrease in RPM.

18. Engine failure below 1500' AGL and 180 KIAS = EJECT (always).

19. Fire on takeoff = Abort, if unable and the fire is confirmed then EJECT.

20. Aircraft Settling off Cat = Throttle at full, maintain 24 units AOA, if settle not stopped then EJECT.

21. Blown Tire on Takeoff = Rudder as req to counter swerve and anti-skid off prior to braking.

22. Fire light in flight = first step is Throttle idle (min for safe flight).

23. Abort procedures (you have to order them from a list).

24. ECA 2 caution light means Throttle <90% above 20k and <95% below 20k.

25. Fire light on ground (know procedures... get out of airplane and don't worry about telling anyone over the radio... no comms in procedure).

26. TP HOT is a CAUTION LIGHT (yellow) and not a WARNING LIGHT (red)... don't get drawn off by any answers where they give you a FIRE light and a TP HOT light and ask which warning light causes you to egress! RTFQ!

27. Always trade airspeed for altitude in EJECT situations.

28. Abort for red lights on takeoff when less than refusal speed.

29. Know Airstart altitudes for Immediate, Assisted, and Windmill (Immediate has no altitude restrictions).

30. OIL PRESS light goes out at approximately 18% on start.

31. Increase altitude for better radio reception (really... it's on there).

32. Structural damage to the airframe may require you to blow the canopy.

ABORT:

THROTTLE IDLE

L BRZ - GEC. CONTINUED 1 HYDRAULIC EXT, MECH RETRACTED

AUG ILLUM - FAILURE, BIT TEST PAUSE OFF

ANK, BLINKING, STUCK PUSHBUTTON - MALFUNCTIONING

ADDITIONAL BUTTON FOR 1 display - Display power reset & all sys

6X

Oil press - min
light

EIA 25 90/95

No airstart - FIRE in FLT

Engine Stall

Comp Stall - ↑ EGT & RPM FF sensor

HYD 1 Fail - what Ind.

Secondary of FIRE

Differential brakes - loose NWS

Blown Tire - Anti-SKID OFF

Clutch - ~~Eng~~ GT, HV

F Press Caution light illuminates 30 sec after GET
fail or if one of the engine switches is not on.

LP pump - ↓ 35 psi

6x

ELLINGTON

✓(A)

With a HYD 2 pressure failure RAT will produce:
2500-3000 psi.

✓(B)

With a confirmed in-flight fire, after setting throttle to OFF and pulling the fuel shutoff there are no more secondary indications... you should: PREPARE FOR A CONTROLLED EJECTION

✓(C)

(assuming no blown tire)
FOR AN ABORTED T/O: DO NOT TURN ANTISKID OFF

(D)

FOR A BLOWN TIRE, ~~STOP~~ PRIOR TO BRAKING YOU SHOULD:
TURN ANTISKID OFF

✓(E)

GTS Fire Warning Light is illuminated because there is an overtemperature condition in the GTS compartment. (not b/c there is a fire in the GTS burner can.) ~~YB~~

(F)

If there is an overspeed condition on start, the first step is: Throttle - OFF

✓(G)

On start, if the engine does not light off 15 seconds after selecting Throttle-IDLE, the pilot should suspect a wet start.

(H)

GTS Fire requires an emergency egress

CANC ILLUM - FAILURE, BIT TEST, PAUSE OFF

BLANK, BLINKING, STUCK PUSH BUTTON - MALFUNCTIONS

INDIVIDUAL BUTTON FOR 1 display - Display power reset for all sys

"PRESS"

E.P. GOUGE

Two 40 question tests: this gouge is what we could remember in no particular order.

1. Know cabin alt. warning light procedures. ① GROSS - ONLY MASK TIGHT ④ ALT - DESCEND. BELOW 25,000' MSL
② REDUCE AOA
③ AIRFLOW METER - OFF THEN ON
- (COUNCIL AT REMAINS ON)
2. Total electrical failure procedures (you don't have to do load shedding because it's a total failure.)
3. Trim failure question: answer ADI indicates 2 deg. nose up.
4. Directional control in question: arrestment is 50 ft. prior to the gear.
5. Coming in for the break and you get slats light, slow to and maintain below 200 KIAS.
6. Something about a hyd fail with RAT extended but the aircraft is uncontrollable. The key word is "uncontrollable." EJECT! ... you bet.
7. Engine stall: throttle _ idle, controls _ neutral, EGT/RPM _ monitor
8. READY light = N1 100 RPM in correct direction and ignitors energized
9. When you have a midair with your wingman, the answer is have someone else give you a visual inspection, not the guy you just bumped into.
10. Power setting for oil press light: idle.
11. MFD FAIL: blinking screen, stuck push button, ?
12. Engine failure off cat: EJECT. If unable to eject off cat: ditch straight ahead
13. TP HOT lights on deck: get out! Same exact procedures as fire on deck.
14. Wheel brake failure on touchdown: go around! If unable to go around: feet off brakes, ANTI SKID _ off, reattempt braking
15. Canopy loss: descend to 10K or min. safe alt.
16. Question about pitot static failure: answer stby vsi
17. Question about launch bar landing: true



* KNOW THE BOLD FACE

* ECAZ CAUTION LIGHT — ① Do not exceed 90% T. 20,000' or
Do not exceed 95% ↓ 20,000, Do not exc
600°C EGT

* FIRE WARNING LIGHT — DO NOT DO AIRSTART, PREPARE FOR EJECT

* SETTLING OFF CATAPULT — MAINTAIN 24 UNITS AOA

* ENGINE FAILURE DURING CATAPULT — ① EJECT ^{NOT POSSIBLE}
② DITCH STRAIT AHEAD

* HYD Z FAILURE — HYD 1 PRESSURE — MONITOR (NOT ACCUMULATOR)

* STALL : FIRST STEP ① TROTTLER IDLE (BANGS, EGT ↑, RPM ↓)
- VS -

FLAMEOUT — FIRST STEP ① TROTTLER OFF

STALLS — DO YOU PERFORM AIRSTART — NO } ① First
DO YOU PULL EMERG OZ — NO }

FLAMEOUT — YOU DO THOSE FIRST

* HYD 1 FAILURE — MANUAL GEAR, EMERGENCY FLAPS, ARRESTED LANDING

"WHAT DO YOU HAVE AVAILABLE, YOU DONT HAVE GENERAL SERVICE"
ANSWER : ALL THE ABOVE

ELECTRICAL FAILURE — KNOW WHAT ALSO NOT AVAILABLE "FLAPS / EMERG" NO

* SECONDARY INDICATIONS OF FIRE : EGT ↑, FUEL FLOW ↑, FIRE LIGHT

* GTS FIRE FIRST STEP : ~~TROTTLER~~ ① ENGINE SWITCH — OFF
② TROTTLER — OFF

* OVERSPEED START — TROTTLER OFF

* NO NWS — USE DIFFERENTIAL BRAKING

* EJECTION — CONVERT AIRSPEED TO ALTITUDE, IF FLIGHT CONTROLS UN-
ABLE
EJECT

* DONT HAVE WINGMAN JOINUP — AFTER MIDAIR

* BLOWN TIRE — ANTI-SKID OFF (ON T/O or LANDING)

* DITCHING — GEAR ↑, FLAPS ↓

* IMMEDIATE AIRSTART — ON ANY ENGINE FAILURE!! (except ↓ 1500 < 180

18. Don't worry about PA procedures
19. Cabin Alt. Light: descend below 25K
20. Engine failure below 1500 AGL and 180 KIAS: EJECT
21. Fire on t/o: if able: abort, if unable and fire is confirmed: EJECT
22. A/C settling off cat: check throttle full fwd, maintain 24 units AOA, if settle not stopped: EJECT
23. Blown tire on t/o: rudder_ as req'd to counter swerve, ANTI-SKID_off prior to braking
24. Know smoke/fumes procedure
 - ① ALT - DESC. < 25,000' (if practical)
 - ② AIRFLOW knob - OFF
 - IF UNABLE TO CLEAR SMOKE -
 - ③ A/C REDUCE
 - ④ WHEN OCCUPANT/SECURE SEAT
 - ⑤ SEAT ↓ ⑥ VIBR ↓ ⑦ MDC - ALL ⑧ CLIMB AS POSSIBLE
25. Dual blown tire landing: (pretend you're an Airforce weenie and) FLARE the landing
26. OBOGS failure: if it won't reset, leave OBOGS OFF. Use the green apple - you don't want bad O2 from OBOGS contaminating good emergency O2.
27. Fire in flight questions
28. Generator failure
29. GTS fire in flight
30. Hook light means hook not down with handle down
31. Question about gear not being down. Answer: warning light in the handle
32. Know the 4 things you lose with C AUG caution light: yaw damper, SBI, ARI, rudder trim.
33. Question about ECA fail: know that it's 90% above 20K and 95% below 20K.
34. CABIN ALT light: OBOGS_on, mask_tight, AOA_reduce, airflow_off

MEMORIZE

- ① FIRE
- ② LTS FIRE
- ③ GENERATOR FAILURE

- * CAUTION Light Lose - ARI, SBI, YAW AUG, RUDDER TRIM
- * Settle off CAT - 24 UNITS AOA
- * WINDMILL $\downarrow$ 25K, 13% min RPM, +250 KIAS
- * GEN FAILURE - V ENGINE RUNNING, Don't hold Reset in Reset on momentarily
- * CABIN ALT (Pressurization failure) * Descend Below 25,000' MSL CABIN ALT CAUTION
OBOGS ON
MASK TIGHT
- * OBOGS - Descend below 10,000' MSL *
- * BLOWN TIRE - NO ARRESTING gear, RUNOUT OPTION, FIRED LANDING, $\frac{1}{2}$ Flt Single
- * DUAL BLOWN TIRES - Gear up or eject if Arresting gear N/A
- * KNOW ELECTRICAL FAILURE VERBATUM
- * ~~ARRESTING GEAR~~ ^{SHORT FIELD} ~~500' - 1000' prior~~ ^{PRIOR}
- * 2° ADI PITCH DURING EXT or retraction of BOARDS NORMAL w/ SBI
- * LOAD SHEDDING - NO FACTOR WITH TOTAL ELECTRICAL FAILURE
- * KNOW SLAT Light positions
- * LEARN ALL CAUTION LIGHT EP'S VERBATUM (Kiddling)
- * IF YOU DO THE CAUTION LIGHTS VERBATUM YOU'RE A JACKASS!
- S/B BLOW BACK CAN OCCUR @ 380, may not extend all the way 340.
- SHORT FIELD & LONG FIELD ARRESTMENTS

*

EC

1. CABIN ALT Warning Light Procedures: *CABIN PRESSURE OR A/C FAILURE*
↳ DOESN'T MEAN YOU LOSE O₂
- OBOGS – on and mask TIGHT
 - Reduce AOA (if required)
 - AIR FLOW Knob – OFF, then ON

If Warning Light remains on –

- Altitude – DESCEND BELOW 25,000' MSL
- Land as soon as PRACTICAL

2. TOTAL ELECT. FAILURE procedures:

- Emergency oxygen actuator – PULL
- Proceed to VMC as soon as POSSIBLE
- Land as soon as PRACTICAL
- Attempt to reset generator / check Batt Switches ON
- Landing Gear – EMER. EXTEND
- Plan a NO FLAP / SLAT approach and arrested landing, if available.
 Recommend a short field arrestment due to no NWS

3. Trim Failure question – answer will say ADI indicates 2 deg. Nose up

4. MFD FAILURE: Indicated by

- blank screen
- blinking screen
- stuck button
- multiple anomalies

5. WHEEL BRAKE FAILURE ON TOUCHDOWN:

- Go around, if able

If go around not feasible-

- Wheel brakes – release
- ANTI-SKID – OFF
- Attempt braking
- Hook – DOWN (1,000' prior to gear)
- Throttle – OFF (it depends how)

If airborne –

- ANTI-SKID – OFF
- Attempt a short field arrestment

6. CANOPY LOSS: descend to 10,000

7. OBOGS Contamination (OXYGEN Warning Light): descend to less than 10,000
 CABIN ALT or Minimum Safe Altitude

1 IN FAILURE → VOR ILS MB RNDALT (LOST)

burn FALL → TRN ~~LOST~~ AZIMUTH (LOST)

APP AOA = 17 UNITS 160 KIAS (APPROX)

4/D MAX = 14 UNITS 180 KIAS (APPROX)

CRUISE = 12 UNITS 230 KIAS (APPROX)

EMERG GEAR HANDLE → DISABLES NWS

BROWN TRIP ON TAKEOFF

BROWN TRIP DURING CAT INCH

LANDING W/ BROWN TRIP(S)

GREEN TRIP ON TAKEOFF

BROWN TRIP ON TAKEOFF

GREEN TRIP ON TAKEOFF

REVIEW ECS

9. Question about PITOT STATIC FAILURE: answer - STBY VSI

10. CABIN ALT LIGHT: descend below 25K (BECAUSE YOU DON'T LOSE O₂)

11. BLOWN TIRE ON I/O:

If aborting

- Rudder - as required to counter swerve
- ANTI-SKID switch - OFF PRIOR TO BRAKING

If taking off

- Rudder - as required to counter swerve
- Gear and Flaps - DO NOT REPOSITION
- Eng. Instruments - MONITOR
- Execute Landing w/ Blown Tire procedure

12. SMOKE OR FUMES FROM COCKPIT

- Altitude - Descend Below 25,000' (if practical) *You make the call*
- AIR FLOW knob - OFF

If unable to clear smoke or unable to see -

- Airspeed - REDUCE
- Warn other cockpit occupant / secure loose items
- Seat - LOWER
- Visor - DOWN
- MDC firing handle - PUL
- Land as soon as POSSIBLE

13. For a DUAL BLOWN TIRE LANDING - FLARE the Landing (*Consider gear up or controlled glider*)

14. OBOGS FAILURE - if it won't reset, leave OBOGS - OFF. Use the green apple, you don't want bad O₂ from OBOGS contamination good emergency O₂

15. GENERATOR FAILURE

- Confirm Engine Operation
- GEN switch - RESET
- VOLTMETER - CHECK

If Generator resets and voltage is abnormal or generator does not reset

- BATT switches - CHECK ON
- GEN switch - OFF
- Unnecessary electrical equip. - SECURE
- Proceed to VMC
- Land as soon as PRACTICAL

If MFD's required

- DISPLAY POWER - ORIDE
- DISPLAY POWER - Normal, when MFD's no longer needed

- Continue Normal Operations

16. What is the first thing you do when you get a GENERATOR Warning Light?

- Confirm Engine Operation

17. HOOK light means that the hook is ~~NOT~~ down with Handle DOWN
Does not agree w/ handle

18. For the question about the gear not being down, the answer is: Warning Light in the handle

19. What have you lost when you get a C AUG Caution Light?

- Yaw Damper
- SBI *→ SLAP REPOSITION*
- ARI *→ QUICK ROLL CORR*
- Rudder Trim (Auto ~~CLAMP~~ CLAS)

20. SBI does not change stick position

21. Speed Brakes may not fully extend ABOVE 340 KIAS

22. Speed Brake Blowback will occur ABOVE 380 KIAS

23. The slower you go, the more control you have

24. C AUG BIT – for it to work, the FLAPS must be UP, and less than 80 Knots, *!Vc w?*

25. Gear Handle Light – will illuminate when the gear is in transit or any time the Gear is not in the commanded position

26. LONG FIELD ARRESTMENT procedure (for test only)

- Hook – DOWN (1,000' prior)
- Maintain Directional control
- Release brakes as you cross the cable

27. A 2 degree pitch during extension or retraction of the boards is normal with the SBI *if*

28. Load Shedding is not a factor with TOTAL ELECTRICAL FAILURE *RTRQ*

29. SLAT Light – A Caution light that will illuminate if SLATS not in proper position, extended above 217 knots, or split, *SLATS EXTENDED + ACCURATED 217*

30. Coming in for the break and you get the SLATS light, slow to and maintain below 200 KIAS

ORVILLE ARREST

NWS AUG FAILURE ~~AND~~ TD JUST BEFORE (50') ARRESTING GEAR

L-BAR EXTENDED $\rightarrow$ LAND PAST ARREST GEAR

31. For a Landing with a SINGLE BLOWN TIRE and no available arresting gear:

- Perform a flared landing, FLAPS 1/2

32. For a Landing with DUAL BLOWN TIRES, and no available arresting gear:

- Consider gear UP landing, or a controlled ejection

33. SHORT FIELD ARRESTMENT - if directional control NOT in question:

- Touch down 500' prior to gear

If directional control IS in question:

- Touch down at or 50' prior to gear

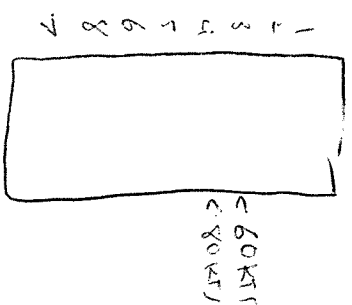
34. Settle off CAT - T handle ✓ full forward

Maximum 24 in AOA

It not stopped

Eject.

35. Windmill Arrestment - to 200 K, 130/100 min A/H, 250 KIAS or greater.



36. Multiple MFO Eject - Deploy gear first

EP 11X

- NC 1. Know Cabin Altitude Warning Light procedures
- OBOGS – on and mask tight
 - Reduce AOA (if req)
 - AIR FLOW KNOB – off then on
- If Warning Light remains on –
- Altitude – Descend Below 25,000' MSL
 - Land as soon as practical
- 1/2001 NC 2. Know Total Electrical Procedures – NO LOAD SHEDDING NECESSARY!
(you can eliminate three of the four answers because they include load shedding)
- Emergency Oxygen Actuator – Pull
 - Proceed to VMC as soon as possible
 - Land as soon as practical
 - Attempt to reset Generator/Check Batteries on
 - Landing gear – Emergency Extend
 - Plan a NO FLAP/NO SLAT approach and arrested landing, if available. Recommend a short field trap due to no NWS
- NC 3. If Directional Control is ever an issue, then take a short field arrestment and land 50' prior to the gear with your hook down.
- NC 4. If you are coming into the break and you get a slats light, slow to and maintain below 200 KIAS.
- ✓ 5. Engine Stall (or compressor stall) first step is Throttle – Idle.
- NC 6. MFD Failures = Blinking Screen, Stuck Button, Black Screen, Wrong Info.
- R NC 7. Wheel Brake Failure on Touchdown = GO AROUND! If unable, then feet off brakes, anti-skid off, attempt braking.
- NC 8. Canopy Loss = Descend to 10k or min safe altitude and Less than 200 KIAS.
- NC 9. Question about Pitot Static Failure = Check STBY VSI.
10. Any Engine Failure below 1500' AGL and 180 KIAS = EJECT.
- 7E 11. Engine Fire on Takeoff = if able, ABORT, if unable and fire confirmed EJECT.
12. Aircraft Settling Off Cat = check throttle full forward, maintain 24 units AOA, if settle not stopped EJECT.

• WHEELS WARNING - <170 KIAS <93 ft. 67200 HSC
• FLAPS 4

13. Know Smoke/Fumes Elimination Procedure

- a. Altitude – Descend below 25,000' (if practical)
- b. Air Flow Know – Off

If unable to clear smoke or unable to see –

- c. Airspeed – Reduce
- d. Warn other cockpit / Secure loose items
- e. Seat – Lower
- f. Visor – Down
- g. MDC Firing Handle – Pull
- h. Land as soon as possible

14. OBOGS Failure – If it won't reset, leave OBOGS OFF.

15. Know Generator Failure procedures

- a. Confirm engine operation
- b. GEN switch – Reset
- c. Voltmeter – Check
- d. Battery Switches – Check On
- e. Generator Switch – Off
- f. Unnecessary electrical equipment – Secure
- g. Proceed to VMC
- h. Land as soon as practical

If MFD's are required –

- i. Display Power Switch – ORIDE
- j. Display Power Switch – Norm when MFD's no longer needed

If Generator resets and voltage is normal –

- k. Continue normal operations

16. Hook Warning Light means hook is not down with handle down.

17. If the landing gear handle is down and you only have two gear down and locked then you will get a light in the gear handle.

18. Know the 4 things you will lose with a C AUG caution light –

- a. Yaw Dampener
- b. SBI
- c. ARI
- d. Rudder Trim

19. OBOGS contamination (Oxygen Warning Light) = Descent to below 10,000 CABIN altitude or Minimum Safe Altitude.

20. If you get a Launch Bar warning light at the boat, you want to divert to a field and make a flared landing after they strip the short field arresting gear.

TEST 1

21. Blown Tire on T/O or Landing

- a. Rudder – As required to counter the swerve
- b. Anti-skid Switch – Off prior to braking

NY

22. For a single blown tire on landing = Flare the landing with ½ flaps.

23. Gear Handle Light will illuminate any time the gear is in transit or any time the gear is not in the commanded position (or the gear doors are not up when the gear is up and locked).

NY

24. Barometric Altimeter Failure = Check Pitot Heat, Use RADALT if less than 5000', Use Cabin Altimeter if unpressurized.

25. Failure of the 28VDC Generator Bus = Loss of all trim except Primary Stab Trim.

NE

26. Windmill Airstart = <25K, Min 13% N2 RPM, Recommended minimum of 250 KIAS.

NY

27. For a Multiple MFD Failure – Display Power Switch to Reset.

NY

28. For a SLATS caution light always maintain below 200 KIAS and land as soon as PRACTICAL (i.e. for a missed approach).

NY

29. Anti-Skid Failure = Brakes Release, Anti-Skid Off, Attempt Braking.

TEST 2

30. Know GTS Fire on deck procedures

- a. Engine switch – Off
- b. Throttle – Off
- c. Fuel Shutoff Handle – Pull
- d. Battery Switches – Off
- e. Egress

TEST 2

31. Know Engine Fire in Flight procedures

- a. Throttle – Idle (min for safe flight)
- b. Secondary Indications – Check
If secondary indications exist –
- c. Emergency Oxygen Actuator – Pull
- d. Throttle – Off
- e. Fuel Shutoff Handle – Pull
If secondary indications persist –
- f. EJECT
If fire is extinguished –
- g. Prepare for controlled ejection
If unable to eject –
- h. Prepare for Flameout Approach

✓ 32. RADALT is operative below 5,000 feet.

↑↑ 33. Know Immediate Airstart Procedures

- a. Throttle – Off
- Simultaneously perform steps 2 and 3
- b. GTS – Press and Hold
- c. Throttle – Idle

↑↑ 34. Know Tailpipe Hot on deck caution light procedures

ENG Switch
OFF

- a. Throttle – Off
- b. Fuel Shutoff Handle – Pull
- c. Battery Switches – Off
- d. Egress

↑↑ 35. Know Engine Fire on deck procedures

- a. Throttle – Off
- b. Fuel Shutoff Handle – Pull
- c. Battery Switches – Off
- d. Egress

✓ 36. The first step of the Clear Engine Procedure is Throttle Off.

Flight Tested

- A V E F A M E

Page 1 of 2

is [FBI]

١٠٠٠
 ١٠٠٠

70

2.

first course —
head of the
racing point
30° beyond —

27 BDAV

Entered course
end of the course
2. # of days —

光緒十一年
正月

2 x 7 from
and a fix.
parallel

150

in & having
time to do so

DIMES + 45²
HEND

$$P_C \rightarrow P_{\text{min}}$$
$$0.15 \times \frac{60}{100} = 0.09$$

radial → $\frac{1}{r}$

1 1/2 oz for lead

Head Tails / Tail

... ..
... ..

Non-free
but all

NLT 2000

NLT 200 1/2

T/O mins

7

$00 - \text{male}$ $3000-3$ or -7
 $\text{male} - 3000-3$ NP P
 $3000-3$ $\text{Male} + 3000-1$ $\text{male} + 2000-10$
 NP/A

20. What are the additional reports made to ATC without special request when not in radar contact:
 - a. When leaving the FAF inbound (non-precision), when leaving the outer marker or fix inbound (precision)
 - b. Corrected estimate any time it becomes apparent an estimate as previously submitted is in error in excess of 3 minutes
 - c. When encountering weather conditions which have not been forecast, or hazardous conditions which have been forecast
 - d. Answer: All Of The Above
21. FSS Freq:
 - a. 255.4
22. Can OFFICIAL BUSINESS ONLY airfield still be used as an alternate
 - a. YES
23. What are the fuel reserve requirements for Jet Log/DD-175
 - a. The greater of 10% of planned requirements or 20 min at ^{WAT} endurance at 10,000' (they try to trick you with 20%.....or 10 min)
24. Be able to calculate arc to radial lead points and vice versa:
 - a. Arc to radial (60/arc nm X 1%GS)
 - b. Radial to arc (1% GS)
25. For the Maxwell alternate question look at all the approaches for lowest non-precision mins (I think the glideslope is NOTAM'd out on this question)
 - a. 700-1 1/2 (400 1/2 + 300-1) *you HAVE TO LOOK AT LOW CHARTS*
26. For the test
 - a. Ensure you read the questions carefully paying particular attention to any NOTAMs or other information that will change the approach minimums
 - b. Remember that ASR minimums are found at the front of the approach plates (There may have been a question where you had to look one up)
 - c. On the JET LOG: ensure you fill in all the spaces (including emergency fields [1 for every 3rd of your route])
 - d. Tab these airfields (you will be looking at one or more approaches at each)
 - i. Meridian NAS (be carefull not to get confused with meridian/key field)
 - ii. Maxwell AFB
 - iii. Jacksonville NAS, FL

27. Holding at Piquay - adjust holding to commence approach at (1610?) you know holding alt + speeds

AT 310

NAVY 230

TACAN ✓ ± 4%

MSA - 1,000 man made + terrain

ESA - 1,000 man made, 2,000 terrain

When up + over glide path. What IS FAF.

Direct lat routes can't be greater than 260 (130) ^{each}

→ DCA
 Wilco → Service
 Standard → Service
 100.0 → 117.95
 1500 → 15
 DR Radial
 Negotiable
 Comp

-

7-4th of designated Radiol
1/2 mi or 3% of the distance to the facility }

(1) SOLD
(2) NIGHT / 1/2
(3) THURSDAY A/C

— Lowest N/Precision: 200% or Taken wires Not allowed to fly through (WV) except if shown that the 300-1

- 1) Emergency
- 2) Hasing developed as planned
- 3) special workshops
- 4) + maintain VMC

- (1) A/c perfrom
- (2) WA
- (3) Route Syta

- Have the following pages tabbed for quick reference
 - NMM 19L Hi-TALAN
 - Maxwell Air Force Base Hi
 - EL Paso Int... ~~19L~~ (South west plate)
 - NOTAM

— Fuel Requirement for DD 175

reserve - 10% of planned requirements or 20 mins of flight
 they'll try to trick you with
 20% ----- or 10 min → don't

Decision Height vs. MDA (see landing minima in App Plate - pg. X in front)
 517 for ILS to 19L at NMM (I think this one is on the form)
 417 for PAR to 19L at NMM

- ILS to Rwy 19L -- NOTAMS indicating ALS inop (yes; you have to ~~write~~ the NOTAM code
 you must increase vis. $\frac{1}{2}$ mile for ILS
 - I don't know if it is the exact runway just look at asterics

Jet Log - climb to 350 (70 nm; 0+12) - still climbing past TAY, Don't

ETE 1:09

Use MXF as Alternate

NMM to MXF is 092°

Spare Fuel 120 to 140 lbs

30° inbound rad intercept = CRS → HD → +30°

DN the L off the bow inbound method = CRS → HD → 30°

Rad into a Rr passage

45° outbound rad int = tail → bow ptr + 45°

FLOW FOR T/O

"McCain Twr, Navy 1A123, #1 at the hold short, ready for takeoff."

"Navy 1A123, McCain Twr, winds are 340 at 10kts cleared for takeoff, switch departure."

"123, cleared for takeoff r/w 01, switching."

ICS "Groove is clear, t/o checks are complete" *going Hot mike*

MRT checks:

1. align with r/w, pump and hold brakes - *HOT mike*

ICS "Brakes hold, MRT checks."

2. P - MRT

ICS "No overspeed, No overtemp, No Warnings, No Cautions, Good volts, (wipeout crtl) Good hyd's, time is __, ready to roll."

Release brakes, start t/o roll with fwd stick. tech - make only 2° corrections for hdg ctrl.

When a/s is > 50kts off the ADI...

ICS "a/s is alive, 90kts (before or at the gear, otherwise abort), good jet going flying, 120 rotate"

N/U 10°, do not exceed 17 units

@ 100' AGL (or 400' MSL @ NMM)

ICS "gear"

ICS fm IP "clear"

Raise the gear

@ 300' AGL (or 600' MSL @NMM) & > 140kts

ICS "flaps"

ICS fm IP "clear"

Raise the flaps

ICS "clean before 200kts, handle checked (gear handle)" *Going Cold*

"Meridian Departure, Navy 1A123, passing _ thousand _ hundred for _ thousand."

Only time to use hundreds of feet when talking to ATC.

Accel and when approaching 240-245kts, N/U another 7° for ~ total of 17° N/U to maintain a 250kts climb out.

Maintain 250 kts below 10k'.

BI stage - Maintain 250kts in the climb even above 10k'

RI stage - Above 10k', N/D 3°, accel to & maintain 300kts until intercepting 0.72IMN (at ~25k')

Remember if your initial lvl/o is > 10k' do the 10k'/15min. checks, obviously over ICS to IP

- | | |
|---------------------------|----------------------|
| 1. Flt & Eng. Instruments | check |
| 2. Cabin Press | check |
| 3. Fuel State | __ lbs. |
| 4. LAW | check, set for 5000' |
| 5. Location/Navaid setup | check |

Begin lvl/o at the lead pt, 10% of VVI/VSI prior to lvl/o altd, by simult.:

- P) 1200-1400 pph (cruise @ 250kts)
A) adj. nose attd for lvl flt
T) trim

Lvl/o check of airspeed, altd and clock

ICS "one five thousand, mark; two fifty, mark"

ICS "standby to mark the clock: three, two, one, mark, ..., mark time is 5 secs"

FLOW FOR DEPARTURE

Use 1% of g/s (displayed below ownship on HSI) to determine when to turn onto the arc (most arcs @ NM/M are 10DME).

Gouge - turn nlt 7.5DME outbd onto a 10DME arc, you are probably late and will overshoot (hint: tell the instructor you will need > 30° AOB to make it)

Gouge - Turn off a 10DME arc at 13 radials prior.

Pine Hill Dept - LRs for 161° rad are either 148° or 174°

Dancy Dept - LRs for 007° are either 354° or 020°

"Navy 1A123, Memphis Cntr, cleared into Pine Hill West or Area 4, monitor btn 11 or btn 8. There are other a/c in the area."

"Memphis Cntr, Navy 1A123, cleared Pine Hill West or Area 4, switching."

"99, Pine Hill West or Area 4, 123 checking in from the North."

Gouge - 17.5k' xsit altd for Area 4.

FLOW FOR RECOVERY/APPROACHES

Bl stage - instructor will give a vector and altd and ask you to check out of the working area.

"99, Pine Hill West, 123 checking out for a Key Field recovery."

Switch to btn 9, ATL Cntr.

"Atl cntr, Navy 1A123, checking out of Pine Hill West for a Key Field recovery."

"Navy 1A123, Atl Cntr, cleared for the Key Field recovery."

Depending on distance and traffic....

"Navy 1A123, Atl Cntr, contact Meridian App, local btn 15."

"Atl Cntr, Navy 1A123, switching btn 15."

Gouge - If ATIS has not been checked, GET IT NOW prior to switching up App Cntl. (Hint: use back radio to get ATIS) (also, Key Field ATIS freq is found on the airport diagram in the high app plates)

"Meridian App, Navy 1A123, level __, request."

"Navy 1A123, Meridian App, go ahead with your request."

"Meridian App, Navy 1A123, req. a HI-TCN key field to a PAR at McCain." Information XH

Xsit to IAF, do the descent/pen checklist and put TWR freq in back radio.

@ 35DME

P - Idle, bds out

A - 10° N/D

T - trim

Leaving AH

"Meridian App, Navy 1A123, out of one six thousand on penetration for HI-TCN B app at Key."

"Navy 1A123, Meridian App, cleared for the HI-TCN app at Key Field. Map instructions are..."

"Meridian App, Navy 1A123, cleared for the HI-TCN app at Key, (repeat MAP instructions)."

If you have not negotiated for a follow on app at McCain be sure to do so at the end of the above xmission.

1000' prior to lvl/o, bds in

@ 5300' MSL (or 5000' AG)

ICS "Platform"

P - adj. if needed, bds in

A - adj. pitch to cut descent rate in 1/2

T - trim

Use 1% of ground speed to figure lead point to turn onto arc.

Follow the rest of the approach to turn onto final radial of 320° using 1% of ground speed times the # of radials/nm for the arc distance.

Once you are on the final radial slow to 200 KIAS and call for the gear.

Do the landing checks and slow to on-speed.

@ the FAF: Time (start the clock), turn (N/R), time (N/R), transition (BDS out pwr back a little for the descent and say "bds out, landing checklists complete", twist (N/R), talk ("Meridian App, Navy 1A123, FAF, three down and locked."

Fly to MDA and then execute a MA and the MAP.

07/10/02

BI TEST GOUGE

(in no particular order but deadly accurate)

30°	10sec	150° 50sec
60°	20sec	180° 1min
90°	30sec	
120°	40sec	

1. SRT - 3° per second (30°/10 sec)

Airspeed does not matter...SRT at 140 kts is still 3° per second

2. ½ SRT - 1 ½° per second (30°/20 sec)

How long will it take to turn 210°? 140 seconds

30°	20sec	120° 1m 20s
60°	40sec	150° 1m 40s
90°	60sec	180° 2min

3. Turns of 30° or less amount of turn equals angle of bank

If you need to make a turn of 15° of heading change...use 15° AOB

4. Turn pattern 2 * 30° for 60° heading change 2 * 45° for 90° heading change

5. S-1, S-3 - descent 700 pph, climb 1500 pph

6. S-1, S-3 - 3 sec lead or 100 ft whichever comes first

7. During an S-3 reverse ½ SRT at beginning of second descent

LOW TERMINAL HL TACAN

INBOUND 30°

DOUBLE 7 OFF BON

8. When recovering from a dirty stall maintain 23-24 units AOA

OUTBOUND 45°

9. UA nose high recovery - Neutralize, analyze, MRT, speed brakes in, 5-10 units AOA
Upright - 19 units / Inverted - 5-10

10. Altitude loss while performing Split-S - 6000ft

11. Extend boards if airspeed reaches 300 kts during nose low recovery

12. Begin loop at 380 kts and approx. 96% RPM

13. Radial intercepts: one 45° intercept and one 30° intercept
The answers are 135 and 220 (can't remember which is for what)

135 - 45° int

220 - 30° int

14. Lose VHF? VOR/ILS indications disappear (there is no off flag)

15. Lose GINA? Lose waypoint, time-to-go, wind direction and groundspeed (no GINA advisory on MFD)

Any info + position info
BLANKING SENSE ADI, HDING
NET FREEZING

16. Groundspeed Calculation - 11 miles in 2 minutes speed = 330

60.0
..9.5-

1m = X 60

2m = X 30

3m = X 20

17. Arc to radial - 60/DME x 1% of groundspeed

At 30 DME and 200 kts your lead will be 4 radials

31.1
33.5.0

18. Radial to arc - 1% groundspeed (GS not TAS)

300 KGS

½ SRT

60 x 10% GS = Lead Radial

1 of 2

(643) 437-7145 (0...)